Theory of Mathematics

Volume IV: Complex Analysis and Riemann Surfaces

Contents

Part I

Holomorphic Functions

Chapter 1

Complex Differentiability

Complex differentiability is much more rigid than real differentiability. A single complex derivative must approximate increments from every direction while respecting multiplication by i, and this rigidity quickly leads to analyticity.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

1.1 Complex derivative

A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions.

1.2 Real-linear viewpoint

Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number.

1.3 Directional consistency

Approaching along real and imaginary directions gives necessary compatibility conditions.

1.4 Holomorphic functions

A function holomorphic on an open set is complex differentiable at every point there.

1.5 Elementary examples

Polynomials and rational functions away from poles are holomorphic; conjugation and modulus fail complex differentiability except at special points.

1.6 Algebra rules

Sums, products, quotients, and compositions obey familiar derivative rules where defined.

1.7 Local linearization

$f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|)$ is the coordinate-free local approximation.

1.8 Rigidity preview

Holomorphicity will imply power-series expansions, infinite differentiability, and strong uniqueness principles.

1.9 Core structural results

Theorem 1.1 (Complex-linear characterization)**.** *A real-differentiable map $f : \mathbb{C} \to \mathbb{C}$ is complex differentiable at z_0 iff its real derivative commutes with multiplication by i.*

Proof. If Df is multiplication by $f'(z_0)$ it clearly commutes with i. Conversely any real-linear map commuting with i is determined by its value at 1 and equals complex multiplication by that value. ☐

Theorem 1.2 (Product rule)**.** *If f, g are complex differentiable at z_0, then $(fg)' = f'g + fg'$.*

Proof. Use the increment identity $f(z)g(z) - f_0 g_0 = (f - f_0)g + f_0(g - g_0)$ and divide by $z - z_0$, using continuity implied by differentiability. ☐

Theorem 1.3 (Chain rule)**.** *If f is differentiable at z_0 and g at $f(z_0)$, then $(g \circ f)'(z_0) = g'(f(z_0))f'(z_0)$.*

Proof. Insert the complex-linear first-order expansions for f and g and combine their little-o remainders. ☐

1.10 Worked examples

Example 1.4 (Complex derivative diagnostic)**.** Give a rigorous worked analysis of Complex derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 1.5 (Real-linear viewpoint diagnostic)**.** Give a rigorous worked analysis of Real-linear viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number.

The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 1.6 (Directional consistency diagnostic)**.** Give a rigorous worked analysis of Directional consistency. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Approaching along real and imaginary directions gives necessary compatibility conditions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 1.7 (Holomorphic functions diagnostic)**.** Give a rigorous worked analysis of Holomorphic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function holomorphic on an open set is complex differentiable at every point there. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

1.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 1.8 (Complex derivative diagnostic)**.** Give a rigorous worked analysis of Complex derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.9 (Real-linear viewpoint diagnostic)**.** Give a rigorous worked analysis of Real-linear viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.10 (Directional consistency diagnostic)**.** Give a rigorous worked analysis of Directional consistency. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Approaching along real and imaginary directions gives necessary compatibility conditions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.11 (Holomorphic functions diagnostic)**.** Give a rigorous worked analysis of Holomorphic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function holomorphic on an open set is complex differentiable at every point there. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.12 (Elementary examples diagnostic)**.** Give a rigorous worked analysis of Elementary examples. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Polynomials and rational functions away from poles are holomorphic; conjugation and modulus fail complex differentiability except at special points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.13 (Algebra rules diagnostic)**.** Give a rigorous worked analysis of Algebra rules. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Sums, products, quotients, and compositions obey familiar derivative rules where defined. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.14 (Local linearization diagnostic)**.** Give a rigorous worked analysis of Local linearization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|)$ is the coordinate-free local approximation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.15 (Rigidity preview diagnostic)**.** Give a rigorous worked analysis of Rigidity preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphicity will imply power-series expansions, infinite differentiability, and strong uniqueness principles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 1.16 (Complex-linear characterization proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A real-differentiable map $f : \mathbb{C} \to \mathbb{C}$ is complex differentiable at z_0 iff its real derivative commutes with multiplication by i.

Solution. If Df is multiplication by $f'(z_0)$ it clearly commutes with i. Conversely any real-linear map commuting with i is determined by its value at 1 and equals complex multiplication by that value. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 1.17 (Product rule proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f, g are complex differentiable at z_0, then $(fg)' = f'g + fg'$.

Solution. Use the increment identity $f(z)g(z) - f_0g_0 = (f - f_0)g + f_0(g - g_0)$ and divide by $z - z_0$, using continuity implied by differentiability. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 1.18 (Chain rule proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is differentiable at z_0 and g at $f(z_0)$, then $(g \circ f)'(z_0) = g'(f(z_0))f'(z_0)$.

Solution. Insert the complex-linear first-order expansions for f and g and combine their little-o remainders. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 1.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Complex differentiability is much more rigid than real differentiability. A single complex derivative must approximate increments from every direction while respecting multiplication by i, and this rigidity quickly leads to analyticity. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

1.12 Exercises with complete solutions

Exercise 1.20. State and apply the central principle behind Complex derivative. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A function is complex differentiable at z_0 if the difference quotient has one limit as $z \to z_0$ through arbitrary complex directions. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.21. State and apply the central principle behind Real-linear viewpoint. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Writing $f = u + iv$ identifies the real derivative with a 2×2 matrix; complex differentiability means that matrix is multiplication by one complex number. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.22. State and apply the central principle behind Directional consistency. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Approaching along real and imaginary directions gives necessary compatibility conditions. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.23. State and apply the central principle behind Holomorphic functions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A function holomorphic on an open set is complex differentiable at every point there. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.24. State and apply the central principle behind Elementary examples. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Polynomials and rational functions away from poles are holomorphic; conjugation and modulus fail complex differentiability except at special points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.25. State and apply the central principle behind Algebra rules. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Sums, products, quotients, and compositions obey familiar derivative rules where defined. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.26. State and apply the central principle behind Local linearization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. $f(z_0+h) = f(z_0) + f'(z_0)h + o(|h|)$ is the coordinate-free local approximation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 1.27. State and apply the central principle behind Rigidity preview. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Holomorphicity will imply power-series expansions, infinite differentiability, and strong uniqueness principles. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 2

Cauchy–Riemann Equations

The Cauchy–Riemann equations translate complex differentiability into real partial-derivative relations. They reveal the geometric link between holomorphic maps, conformality, harmonic functions, and complex linearity.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

2.1 Cartesian form

For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$.

2.2 Derivation from directions

Real and imaginary difference quotients must agree, yielding the equations.

2.3 Sufficiency with regularity

If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby.

2.4 Polar form

In polar coordinates, the equations adapt to radial and angular derivatives.

2.5 Harmonic components

Twice differentiable holomorphic functions have harmonic real and imaginary parts.

2.6 Harmonic conjugates

Locally on simply connected regions, harmonic functions admit harmonic conjugates under the appropriate integrability condition.

2.7 Jacobian geometry

The real Jacobian of a holomorphic map with nonzero derivative is a rotation-dilation matrix.

2.8 Failure examples

Functions such as $\bar{z}$ and $|z|^2$ expose why partial derivatives alone do not guarantee holomorphicity.

2.9 Core structural results

Theorem 2.1 (Cauchy–Riemann necessity). *If $f = u + iv$ is complex differentiable at z_0, then $u_x = v_y$ and $u_y = -v_x$ at z_0.*

Proof. Evaluate the difference quotient along real increments and along purely imaginary increments and equate the two complex limits. □

Theorem 2.2 (Sufficiency under C^1 hypotheses). *If u, v have continuous first partials near z_0 satisfying the Cauchy–Riemann equations, then $f = u + iv$ is complex differentiable there.*

Proof. Use the real differentiability expansion and the equations to identify the real derivative matrix with complex multiplication by $u_x + iv_x$. □

Theorem 2.3 (Harmonicity). *If $f = u + iv$ is holomorphic with continuous second partials, then $\Delta u = \Delta v = 0$.*

Proof. Differentiate the Cauchy–Riemann equations and use equality of mixed partials to obtain cancellation in each Laplacian. □

2.10 Worked examples

Example 2.4 (Cartesian form diagnostic). Give a rigorous worked analysis of Cartesian form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 2.5 (Derivation from directions diagnostic). Give a rigorous worked analysis of Derivation from directions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Real and imaginary difference quotients must agree, yielding the equations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 2.6 (Sufficiency with regularity diagnostic). Give a rigorous worked analysis of Sufficiency with regularity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 2.7 (Polar form diagnostic)**.** Give a rigorous worked analysis of Polar form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. In polar coordinates, the equations adapt to radial and angular derivatives. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

2.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 2.8 (Cartesian form diagnostic)**.** Give a rigorous worked analysis of Cartesian form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.9 (Derivation from directions diagnostic)**.** Give a rigorous worked analysis of Derivation from directions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Real and imaginary difference quotients must agree, yielding the equations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.10 (Sufficiency with regularity diagnostic)**.** Give a rigorous worked analysis of Sufficiency with regularity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.11 (Polar form diagnostic)**.** Give a rigorous worked analysis of Polar form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. In polar coordinates, the equations adapt to radial and angular derivatives. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.12 (Harmonic components diagnostic)**.** Give a rigorous worked analysis of Harmonic components. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Twice differentiable holomorphic functions have harmonic real and imaginary parts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.13 (Harmonic conjugates diagnostic)**.** Give a rigorous worked analysis of Harmonic conjugates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Locally on simply connected regions, harmonic functions admit harmonic conjugates under the appropriate integrability condition. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.14 (Jacobian geometry diagnostic)**.** Give a rigorous worked analysis of Jacobian geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The real Jacobian of a holomorphic map with nonzero derivative is a rotation-dilation matrix. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.15 (Failure examples diagnostic)**.** Give a rigorous worked analysis of Failure examples. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Functions such as $\bar{z}$ and $|z|^2$ expose why partial derivatives alone do not guarantee holomorphicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 2.16 (Cauchy–Riemann necessity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f = u + iv$ is complex differentiable at z_0, then $u_x = v_y$ and $u_y = -v_x$ at z_0.

Solution. Evaluate the difference quotient along real increments and along purely imaginary increments and equate the two complex limits. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 2.17 (Sufficiency under C^1 hypotheses proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If u, v have continuous first partials near z_0 satisfying the Cauchy–Riemann equations, then $f = u + iv$ is complex differentiable there.

Solution. Use the real differentiability expansion and the equations to identify the real derivative matrix with complex multiplication by $u_x + iv_x$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 2.18 (Harmonicity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f = u + iv$ is holomorphic with continuous second partials, then $\Delta u = \Delta v = 0$.

Solution. Differentiate the Cauchy–Riemann equations and use equality of mixed partials to obtain cancellation in each Laplacian. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 2.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Cauchy–Riemann equations translate complex differentiability into real partial-derivative relations. They reveal the geometric link between holomorphic maps, conformality, harmonic functions, and complex linearity. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

2.12 Exercises with complete solutions

Exercise 2.20. State and apply the central principle behind Cartesian form. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For $f = u + iv$, complex differentiability suggests $u_x = v_y$ and $u_y = -v_x$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.21. State and apply the central principle behind Derivation from directions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Real and imaginary difference quotients must agree, yielding the equations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.22. State and apply the central principle behind Sufficiency with regularity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. If first partials exist continuously near a point and satisfy Cauchy–Riemann there, then the function is holomorphic nearby. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.23. State and apply the central principle behind Polar form. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. In polar coordinates, the equations adapt to radial and angular derivatives. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.24. State and apply the central principle behind Harmonic components. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Twice differentiable holomorphic functions have harmonic real and imaginary parts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.25. State and apply the central principle behind Harmonic conjugates. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Locally on simply connected regions, harmonic functions admit harmonic conjugates under the appropriate integrability condition. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 2.26. State and apply the central principle behind Jacobian geometry. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The real Jacobian of a holomorphic map with nonzero derivative is a rotation-dilation matrix. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 2.27. State and apply the central principle behind Failure examples. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Functions such as $\bar{z}$ and $|z|^2$ expose why partial derivatives alone do not guarantee holomorphicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 3

Power Series and Analytic Functions

Power series are the local algebra of holomorphic functions. Their radius of convergence controls a disk on which termwise differentiation and integration are valid, and analytic functions inherit remarkable rigidity.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

3.1 Complex power series

A series $\sum a_n(z - z_0)^n$ has a radius of convergence determined by coefficient growth.

3.2 Absolute convergence

Inside the radius the series converges absolutely; outside it diverges.

3.3 Uniform convergence on compact subdisks

On every smaller closed disk, convergence is uniform and normally controlled.

3.4 Termwise differentiation

Differentiated power series has the same radius of convergence and represents the derivative.

3.5 Termwise integration

Primitive series are obtained coefficientwise on disks.

3.6 Analytic functions

A function is analytic when locally represented by a convergent power series.

3.7 Taylor coefficients

For analytic functions, coefficients are derivatives divided by factorials.

3.8 Uniqueness of expansion

A power series representation on a disk is unique because its coefficients are recovered by derivatives at the center.

3.9 Core structural results

Theorem 3.1 (Uniform convergence on compact subdisks)**.** *If a power series has radius R, it converges uniformly on $|z - z_0| \leq r < R$.*

Proof. Choose ρ with $r < \rho < R$. Convergence at radius ρ bounds $|a_n|\rho^n$, giving a geometric majorant for $|a_n|r^n$. ☐

Theorem 3.2 (Termwise differentiation)**.** *Inside the radius of convergence, a power series may be differentiated termwise and the derivative series has the same radius.*

Proof. Use uniform convergence on compact subdisks together with the standard derivative-series convergence criterion, or compare coefficients with the root test. ☐

Theorem 3.3 (Coefficient uniqueness)**.** *If $\sum a_n(z - z_0)^n = 0$ on a neighborhood of z_0, then every $a_n = 0$.*

Proof. Differentiate k times and evaluate at z_0 to get $k!a_k = 0$. ☐

3.10 Worked examples

Example 3.4 (Complex power series diagnostic)**.** Give a rigorous worked analysis of Complex power series. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A series $\sum a_n(z - z_0)^n$ has a radius of convergence determined by coefficient growth. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 3.5 (Absolute convergence diagnostic)**.** Give a rigorous worked analysis of Absolute convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Inside the radius the series converges absolutely; outside it diverges. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 3.6 (Uniform convergence on compact subdisks diagnostic)**.** Give a rigorous worked analysis of Uniform convergence on compact subdisks. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. On every smaller closed disk, convergence is uniform and normally controlled. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 3.7 (Termwise differentiation diagnostic)**.** Give a rigorous worked analysis of Termwise differentiation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Differentiated power series has the same radius of convergence and represents the derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

3.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 3.8 (Complex power series diagnostic)**.** Give a rigorous worked analysis of Complex power series. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A series $\sum a_n(z-z_0)^n$ has a radius of convergence determined by coefficient growth. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.9 (Absolute convergence diagnostic)**.** Give a rigorous worked analysis of Absolute convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Inside the radius the series converges absolutely; outside it diverges. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.10 (Uniform convergence on compact subdisks diagnostic)**.** Give a rigorous worked analysis of Uniform convergence on compact subdisks. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On every smaller closed disk, convergence is uniform and normally controlled. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.11 (Termwise differentiation diagnostic)**.** Give a rigorous worked analysis of Termwise differentiation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiated power series has the same radius of convergence and represents the derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.12 (Termwise integration diagnostic)**.** Give a rigorous worked analysis of Termwise integration. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Primitive series are obtained coefficientwise on disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.13 (Analytic functions diagnostic)**.** Give a rigorous worked analysis of Analytic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function is analytic when locally represented by a convergent power series. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.14 (Taylor coefficients diagnostic)**.** Give a rigorous worked analysis of Taylor coefficients. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For analytic functions, coefficients are derivatives divided by factorials. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.15 (Uniqueness of expansion diagnostic)**.** Give a rigorous worked analysis of Uniqueness of expansion. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A power series representation on a disk is unique because its coefficients are recovered by derivatives at the center. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 3.16 (Uniform convergence on compact subdisks proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If a power series has radius R, it converges uniformly on $|z - z_0| \le r < R$.

Solution. Choose ρ with $r < \rho < R$. Convergence at radius ρ bounds $|a_n|\rho^n$, giving a geometric majorant for $|a_n|r^n$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 3.17 (Termwise differentiation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Inside the radius of convergence, a power series may be differentiated termwise and the derivative series has the same radius.

Solution. Use uniform convergence on compact subdisks together with the standard derivative-series convergence criterion, or compare coefficients with the root test. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 3.18 (Coefficient uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $\sum a_n(z - z_0)^n = 0$ on a neighborhood of z_0, then every $a_n = 0$.

Solution. Differentiate k times and evaluate at z_0 to get $k!a_k = 0$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 3.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Power series are the local algebra of holomorphic functions. Their radius of convergence controls a disk on which termwise differentiation and integration are valid, and analytic functions inherit remarkable rigidity. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

3.12 Exercises with complete solutions

Exercise 3.20. State and apply the central principle behind Complex power series. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A series $\sum a_n(z - z_0)^n$ has a radius of convergence determined by coefficient growth. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.21. State and apply the central principle behind Absolute convergence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Inside the radius the series converges absolutely; outside it diverges. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.22. State and apply the central principle behind Uniform convergence on compact subdisks. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. On every smaller closed disk, convergence is uniform and normally controlled. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.23. State and apply the central principle behind Termwise differentiation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Differentiated power series has the same radius of convergence and represents the derivative. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.24. State and apply the central principle behind Termwise integration. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Primitive series are obtained coefficientwise on disks. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.25. State and apply the central principle behind Analytic functions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A function is analytic when locally represented by a convergent power series. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.26. State and apply the central principle behind Taylor coefficients. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For analytic functions, coefficients are derivatives divided by factorials. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 3.27. State and apply the central principle behind Uniqueness of expansion. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A power series representation on a disk is unique because its coefficients are recovered by derivatives at the center. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 4

Complex Integration

Complex integration integrates a complex-valued function along an oriented curve. Parameterization, estimates, primitives, and deformation ideas prepare the way for Cauchy's theorem.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

4.1 Parametrized curves

A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data.

4.2 Contour integral

Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$.

4.3 Reparameterization

Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign.

4.4 ML estimate

$|\int_\gamma f dz| \le L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound.

4.5 Primitives

If $F' = f$, then contour integrals of f depend only on endpoints.

4.6 Closed curves

Functions with primitives integrate to zero around every closed curve.

4.7 Polygonal and circular contours

Line segments, circles, and piecewise polygonal paths provide standard computational contours.

4.8 Deformation preview

When holomorphicity holds in the intervening region, later theorems will make integrals invariant under suitable contour deformations.

4.9 Parameter singularities

Care is required when curves pass through poles, branch cuts, or points where the integrand is not defined.

4.10 Core structural results

Theorem 4.1 (Reparameterization invariance)**.** *An orientation-preserving C^1 change of parameter leaves the contour integral unchanged.*

Proof. Substitute the new parameter into the real integral and use the chain rule and ordinary change-of-variables formula. □

Theorem 4.2 (ML inequality)**.** *If f is continuous on a rectifiable curve γ, then $|\int_\gamma f dz| \leq L(\gamma) \max_\gamma |f|$.*

Proof. Take absolute values in the parametrized integral and bound $|f(\gamma(t))|$ by its maximum; the remaining integral of $|\gamma'|$ is the curve length. □

Theorem 4.3 (Fundamental theorem for contour integrals)**.** *If $F' = f$ on a domain containing γ, then $\int_\gamma f dz = F(\gamma(b)) - F(\gamma(a))$.*

Proof. Differentiate $F(\gamma(t))$ by the chain rule and integrate the resulting ordinary derivative over the parameter interval. □

Theorem 4.4 (Primitive criterion on connected domains)**.** *If every closed contour integral of continuous f vanishes in a connected open set, then f has a primitive.*

Proof. Fix a base point and define $F(z)$ by integrating f along any path from the base point to z; closed-integral vanishing gives path independence, and a short terminal segment proves $F' = f$. □

4.11 Worked examples

Example 4.5 (Parametrized curves diagnostic)**.** Give a rigorous worked analysis of Parametrized curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 4.6 (Contour integral diagnostic)**.** Give a rigorous worked analysis of Contour integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 4.7 (Reparameterization diagnostic)**.** Give a rigorous worked analysis of Reparameterization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 4.8 (ML estimate diagnostic)**.** Give a rigorous worked analysis of ML estimate. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. $|\int_\gamma f dz| \le L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

4.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 4.9 (Parametrized curves diagnostic)**.** Give a rigorous worked analysis of Parametrized curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 4.10 (Contour integral diagnostic)**.** Give a rigorous worked analysis of Contour integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 4.11 (Reparameterization diagnostic)**.** Give a rigorous worked analysis of Reparameterization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 4.12 (ML estimate diagnostic)**.** Give a rigorous worked analysis of ML estimate. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $|\int_\gamma f dz| \leq L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.13 (Primitives diagnostic)**.** Give a rigorous worked analysis of Primitives. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. If $F' = f$, then contour integrals of f depend only on endpoints. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.14 (Closed curves diagnostic)**.** Give a rigorous worked analysis of Closed curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Functions with primitives integrate to zero around every closed curve. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.15 (Polygonal and circular contours diagnostic)**.** Give a rigorous worked analysis of Polygonal and circular contours. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Line segments, circles, and piecewise polygonal paths provide standard computational contours. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 4.16 (Reparameterization invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: An orientation-preserving C^1 change of parameter leaves the contour integral unchanged.

Solution. Substitute the new parameter into the real integral and use the chain rule and ordinary change-of-variables formula. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.17 (ML inequality proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is continuous on a rectifiable curve γ, then $|\int_\gamma f dz| \leq L(\gamma) \max_\gamma |f|$.

Solution. Take absolute values in the parametrized integral and bound $|f(\gamma(t))|$ by its maximum; the remaining integral of $|\gamma'|$ is the curve length. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.18 (Fundamental theorem for contour integrals proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $F' = f$ on a domain containing γ, then $\int_\gamma f dz = F(\gamma(b)) - F(\gamma(a))$.

Solution. Differentiate $F(\gamma(t))$ by the chain rule and integrate the resulting ordinary derivative over the parameter interval. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 4.19 (Primitive criterion on connected domains proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If every closed contour integral of continuous f vanishes in a connected open set, then f has a primitive.

Solution. Fix a base point and define $F(z)$ by integrating f along any path from the base point to z; closed-integral vanishing gives path independence, and a short terminal segment proves $F' = f$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 4.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Complex integration integrates a complex-valued function along an oriented curve. Parameterization, estimates, primitives, and deformation ideas prepare the way for Cauchy's theorem. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

4.13 Exercises with complete solutions

Exercise 4.21. State and apply the central principle behind Parametrized curves. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A piecewise C^1 curve $\gamma : [a, b] \to \mathbb{C}$ carries orientation and tangent data. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.22. State and apply the central principle behind Contour integral. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Define $\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.23. State and apply the central principle behind Reparameterization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Orientation-preserving reparameterizations leave contour integrals unchanged; reversing orientation changes the sign. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.24. State and apply the central principle behind ML estimate. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. $|\int_\gamma f dz| \leq L(\gamma) \sup_\gamma |f|$ gives the basic quantitative bound. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.25. State and apply the central principle behind Primitives. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. If $F' = f$, then contour integrals of f depend only on endpoints. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.26. State and apply the central principle behind Closed curves. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Functions with primitives integrate to zero around every closed curve. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.27. State and apply the central principle behind Polygonal and circular contours. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Line segments, circles, and piecewise polygonal paths provide standard computational contours. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 4.28. State and apply the central principle behind Deformation preview. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. When holomorphicity holds in the intervening region, later theorems will make integrals invariant under suitable contour deformations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 5

Cauchy's Theorem

Cauchy's theorem is the topological heart of elementary complex analysis: holomorphic functions have zero integral around contractible closed contours. This creates path independence, primitives, and deformation invariance.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

5.1 Triangle theorem

The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error.

5.2 Polygonal domains

Triangulation extends the result to polygonal contours.

5.3 Simply connected domains

On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish.

5.4 Homotopy invariance

Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies.

5.5 Existence of primitives

Zero integrals on closed contours imply primitives.

5.6 Goursat refinement

Continuity of the derivative is not required; complex differentiability alone suffices.

5.7 Multiply connected warning

Loops around holes can carry nonzero integrals even when the function is holomorphic on the punctured domain.

5.8 Topological content

Cauchy's theorem converts local holomorphicity plus global domain topology into global integral identities.

5.9 Core structural results

Theorem 5.1 (Cauchy–Goursat theorem on triangles)**.** *If f is holomorphic on a neighborhood of a triangle and its interior, then the integral around the triangle is zero.*

Proof. Repeatedly subdivide into four similar triangles and choose one carrying at least a quarter of the integral magnitude. Diameters shrink to a point where differentiability gives a linear approximation whose constant and linear parts integrate to zero; the remainder estimate forces the original integral to vanish. ☐

Theorem 5.2 (Primitive on simply connected domains)**.** *Every holomorphic function on a simply connected domain has a primitive.*

Proof. Cauchy's theorem makes integrals around closed contours vanish, so path integration from a base point is well-defined and differentiates back to the integrand. ☐

Theorem 5.3 (Homotopy invariance)**.** *If two closed contours are homotopic through contours inside a domain where f is holomorphic, their integrals agree.*

Proof. Partition the homotopy strip into small cells whose boundary integrals vanish by local Cauchy theory; interior edges cancel, leaving the difference of the two boundary contour integrals. ☐

5.10 Worked examples

Example 5.4 (Triangle theorem diagnostic)**.** Give a rigorous worked analysis of Triangle theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 5.5 (Polygonal domains diagnostic)**.** Give a rigorous worked analysis of Polygonal domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Triangulation extends the result to polygonal contours. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 5.6 (Simply connected domains diagnostic)**.** Give a rigorous worked analysis of Simply connected domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 5.7 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

5.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 5.8 (Triangle theorem diagnostic)**.** Give a rigorous worked analysis of Triangle theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.9 (Polygonal domains diagnostic)**.** Give a rigorous worked analysis of Polygonal domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Triangulation extends the result to polygonal contours. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.10 (Simply connected domains diagnostic)**.** Give a rigorous worked analysis of Simply connected domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.11 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 5.12 (Existence of primitives diagnostic)**.** Give a rigorous worked analysis of Existence of primitives. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Zero integrals on closed contours imply primitives. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 5.13 (Goursat refinement diagnostic)**.** Give a rigorous worked analysis of Goursat refinement. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Continuity of the derivative is not required; complex differentiability alone suffices. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 5.14 (Multiply connected warning diagnostic)**.** Give a rigorous worked analysis of Multiply connected warning. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Loops around holes can carry nonzero integrals even when the function is holomorphic on the punctured domain. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 5.15 (Topological content diagnostic)**.** Give a rigorous worked analysis of Topological content. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Cauchy's theorem converts local holomorphicity plus global domain topology into global integral identities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 5.16 (Cauchy–Goursat theorem on triangles proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic on a neighborhood of a triangle and its interior, then the integral around the triangle is zero.

Solution. Repeatedly subdivide into four similar triangles and choose one carrying at least a quarter of the integral magnitude. Diameters shrink to a point where differentiability gives a linear approximation whose constant and linear parts integrate to zero; the remainder estimate forces the original integral to vanish. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 5.17 (Primitive on simply connected domains proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every holomorphic function on a simply connected domain has a primitive.

Solution. Cauchy's theorem makes integrals around closed contours vanish, so path integration from a base point is well-defined and differentiates back to the integrand. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 5.18 (Homotopy invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two closed contours are homotopic through contours inside a domain where f is holomorphic, their integrals agree.

Solution. Partition the homotopy strip into small cells whose boundary integrals vanish by local Cauchy theory; interior edges cancel, leaving the difference of the two boundary contour integrals. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 5.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Cauchy's theorem is the topological heart of elementary complex analysis: holomorphic functions have zero integral around contractible closed contours. This creates path independence, primitives, and deformation invariance. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

5.12 Exercises with complete solutions

Exercise 5.20. State and apply the central principle behind Triangle theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The theorem can first be proved on triangles, where subdivision and local linearization isolate the holomorphic error. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.21. State and apply the central principle behind Polygonal domains. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Triangulation extends the result to polygonal contours. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.22. State and apply the central principle behind Simply connected domains. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. On simply connected regions, every closed contour is deformable to a point and holomorphic integrals vanish. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.23. State and apply the central principle behind Homotopy invariance. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Contour integrals of holomorphic functions are invariant under suitable fixed-endpoint homotopies. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 5.24. State and apply the central principle behind Existence of primitives. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Zero integrals on closed contours imply primitives. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 5.25. State and apply the central principle behind Goursat refinement. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Continuity of the derivative is not required; complex differentiability alone suffices. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 5.26. State and apply the central principle behind Multiply connected warning. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Loops around holes can carry nonzero integrals even when the function is holomorphic on the punctured domain. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 5.27. State and apply the central principle behind Topological content. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Cauchy's theorem converts local holomorphicity plus global domain topology into global integral identities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 6

Cauchy's Integral Formula

Cauchy's integral formula reconstructs a holomorphic function from boundary values. Its differentiated forms yield analyticity, derivative estimates, Liouville's theorem, and the mean-value property.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

6.1 Integral formula

Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$.

6.2 Circle form

On circles the formula becomes an explicit average weighted by the Cauchy kernel.

6.3 Derivative formulas

Differentiating under the integral gives formulas for every derivative.

6.4 Cauchy estimates

Derivative formulas bound derivatives by boundary maxima and powers of radius.

6.5 Analyticity

Holomorphic functions are automatically represented by convergent Taylor series.

6.6 Mean-value property

Holomorphic functions and their harmonic components satisfy circle-average identities.

6.7 Liouville theorem

A bounded entire function must be constant.

6.8 Fundamental theorem of algebra

Liouville's theorem yields a short proof that every nonconstant complex polynomial has a root.

6.9 Core structural results

Theorem 6.1 (Cauchy integral formula)**.** *If f is holomorphic on and inside a positively oriented simple closed contour Γ and z lies inside, then $f(z) = \frac{1}{2\pi i}\int_\Gamma \frac{f(\zeta)}{\zeta - z}d\zeta$.*

Proof. Subtract $f(z)$ from the numerator so the apparent singularity becomes removable, apply Cauchy's theorem to the regular part, and compute the integral of $(\zeta - z)^{-1}$ around the contour as $2\pi i$. ☐

Theorem 6.2 (Cauchy derivative formula)**.** $f^{(n)}(z) = \frac{n!}{2\pi i}\int_\Gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}}d\zeta$.

Proof. Differentiate the Cauchy formula with respect to z; uniform separation of z from the contour justifies differentiation under the integral. ☐

Theorem 6.3 (Cauchy estimate)**.** *If $|f| \le M$ on $|\zeta - z_0| = R$, then $|f^{(n)}(z_0)| \le n!M/R^n$.*

Proof. Apply the derivative formula on the circle and use the ML inequality with length $2\pi R$ and denominator magnitude R^{n+1}. ☐

Theorem 6.4 (Liouville theorem)**.** *Every bounded entire function is constant.*

Proof. Apply the Cauchy estimate for the first derivative on circles of radius R and let $R \to \infty$, forcing $f' = 0$. ☐

6.10 Worked examples

Example 6.5 (Integral formula diagnostic)**.** Give a rigorous worked analysis of Integral formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 6.6 (Circle form diagnostic)**.** Give a rigorous worked analysis of Circle form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. On circles the formula becomes an explicit average weighted by the Cauchy kernel. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 6.7 (Derivative formulas diagnostic)**.** Give a rigorous worked analysis of Derivative formulas. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Differentiating under the integral gives formulas for every derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 6.8 (Cauchy estimates diagnostic)**.** Give a rigorous worked analysis of Cauchy estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Derivative formulas bound derivatives by boundary maxima and powers of radius. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

6.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 6.9 (Integral formula diagnostic)**.** Give a rigorous worked analysis of Integral formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.10 (Circle form diagnostic)**.** Give a rigorous worked analysis of Circle form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On circles the formula becomes an explicit average weighted by the Cauchy kernel. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.11 (Derivative formulas diagnostic)**.** Give a rigorous worked analysis of Derivative formulas. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiating under the integral gives formulas for every derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.12 (Cauchy estimates diagnostic)**.** Give a rigorous worked analysis of Cauchy estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Derivative formulas bound derivatives by boundary maxima and powers of radius. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.13 (Analyticity diagnostic)**.** Give a rigorous worked analysis of Analyticity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphic functions are automatically represented by convergent Taylor series. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.14 (Mean-value property diagnostic)**.** Give a rigorous worked analysis of Mean-value property. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphic functions and their harmonic components satisfy circle-average identities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.15 (Liouville theorem diagnostic)**.** Give a rigorous worked analysis of Liouville theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A bounded entire function must be constant. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 6.16 (Cauchy integral formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic on and inside a positively oriented simple closed contour Γ and z lies inside, then $f(z) = \frac{1}{2\pi i}\int_\Gamma \frac{f(\zeta)}{\zeta - z}d\zeta$.

Solution. Subtract $f(z)$ from the numerator so the apparent singularity becomes removable, apply Cauchy's theorem to the regular part, and compute the integral of $(\zeta - z)^{-1}$ around the contour as $2\pi i$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 6.17 (Cauchy derivative formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: $f^{(n)}(z) = \frac{n!}{2\pi i}\int_\Gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}}d\zeta$.

Solution. Differentiate the Cauchy formula with respect to z; uniform separation of z from the contour justifies differentiation under the integral. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 6.18 (Cauchy estimate proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $|f| \le M$ on $|\zeta - z_0| = R$, then $|f^{(n)}(z_0)| \le n!M/R^n$.

Solution. Apply the derivative formula on the circle and use the ML inequality with length $2\pi R$ and denominator magnitude R^{n+1}. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 6.19 (Liouville theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every bounded entire function is constant.

Solution. Apply the Cauchy estimate for the first derivative on circles of radius R and let $R \to \infty$, forcing $f' = 0$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 6.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Cauchy's integral formula reconstructs a holomorphic function from boundary values. Its differentiated forms yield analyticity, derivative estimates, Liouville's theorem, and the mean-value property. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

6.12 Exercises with complete solutions

Exercise 6.21. State and apply the central principle behind Integral formula. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Inside a positively oriented simple closed contour, $f(z)$ equals a boundary integral with kernel $(\zeta - z)^{-1}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.22. State and apply the central principle behind Circle form. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. On circles the formula becomes an explicit average weighted by the Cauchy kernel. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.23. State and apply the central principle behind Derivative formulas. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Differentiating under the integral gives formulas for every derivative. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.24. State and apply the central principle behind Cauchy estimates. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Derivative formulas bound derivatives by boundary maxima and powers of radius. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.25. State and apply the central principle behind Analyticity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Holomorphic functions are automatically represented by convergent Taylor series. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.26. State and apply the central principle behind Mean-value property. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Holomorphic functions and their harmonic components satisfy circle-average identities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.27. State and apply the central principle behind Liouville theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A bounded entire function must be constant. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 6.28. State and apply the central principle behind Fundamental theorem of algebra. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Liouville's theorem yields a short proof that every nonconstant complex polynomial has a root. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Part II

Singularities and Residues

Chapter 7

Zeros and the Identity Theorem

Zeros of holomorphic functions are isolated unless the function vanishes identically. This local factorization yields the identity theorem, uniqueness from accumulation sets, and multiplicity as the correct notion of zero order.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

7.1 Zeros and multiplicity

If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero.

7.2 Local factorization

Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0.

7.3 Isolated zeros

A nonzero holomorphic function has isolated zeros.

7.4 Identity theorem

Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain.

7.5 Uniqueness of analytic continuation

A holomorphic continuation, when it exists on a connected overlap, is unique.

7.6 Zero sets

Nontrivial holomorphic zero sets in one complex variable are discrete.

7.7 Multiplicity under products

Orders of zeros add under multiplication and subtract under quotients when the denominator order is smaller.

7.8 Applications

Functional identities, symmetry extensions, and polynomial root counts all use the identity principle.

7.9 Core structural results

Theorem 7.1 (Local zero factorization)**.** *If f is holomorphic near z_0, not identically zero, and $f(z_0) = 0$, then $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$ for a unique integer $m \geq 1$.*

Proof. Take the first nonzero Taylor coefficient a_m and factor $(z - z_0)^m$ from the Taylor series; the remaining power series defines g and has nonzero value a_m at the center. □

Theorem 7.2 (Isolated zeros)**.** *Zeros of a nonzero holomorphic function are isolated.*

Proof. The local factorization has nonvanishing factor g near z_0, so the only nearby zero is the explicit factor z_0. □

Theorem 7.3 (Identity theorem)**.** *If two holomorphic functions on a connected domain agree on a set with an accumulation point in the domain, then they agree everywhere.*

Proof. Their difference is holomorphic and has a nonisolated zero. The isolated-zero theorem forces the difference to vanish identically on the connected domain. □

7.10 Worked examples

Example 7.4 (Zeros and multiplicity diagnostic)**.** Give a rigorous worked analysis of Zeros and multiplicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 7.5 (Local factorization diagnostic)**.** Give a rigorous worked analysis of Local factorization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 7.6 (Isolated zeros diagnostic)**.** Give a rigorous worked analysis of Isolated zeros. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A nonzero holomorphic function has isolated zeros. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 7.7 (Identity theorem diagnostic)**.** Give a rigorous worked analysis of Identity theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

7.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 7.8 (Zeros and multiplicity diagnostic)**.** Give a rigorous worked analysis of Zeros and multiplicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 7.9 (Local factorization diagnostic)**.** Give a rigorous worked analysis of Local factorization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 7.10 (Isolated zeros diagnostic)**.** Give a rigorous worked analysis of Isolated zeros. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A nonzero holomorphic function has isolated zeros. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 7.11 (Identity theorem diagnostic)**.** Give a rigorous worked analysis of Identity theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 7.12 (Uniqueness of analytic continuation diagnostic)**.** Give a rigorous worked analysis of Uniqueness of analytic continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A holomorphic continuation, when it exists on a connected overlap, is unique. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 7.13 (Zero sets diagnostic)**.** Give a rigorous worked analysis of Zero sets. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Nontrivial holomorphic zero sets in one complex variable are discrete. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.14 (Multiplicity under products diagnostic)**.** Give a rigorous worked analysis of Multiplicity under products. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Orders of zeros add under multiplication and subtract under quotients when the denominator order is smaller. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.15 (Applications diagnostic)**.** Give a rigorous worked analysis of Applications. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Functional identities, symmetry extensions, and polynomial root counts all use the identity principle. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 7.16 (Local zero factorization proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic near z_0, not identically zero, and $f(z_0) = 0$, then $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$ for a unique integer $m \geq 1$.

Solution. Take the first nonzero Taylor coefficient a_m and factor $(z - z_0)^m$ from the Taylor series; the remaining power series defines g and has nonzero value a_m at the center. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 7.17 (Isolated zeros proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Zeros of a nonzero holomorphic function are isolated.

Solution. The local factorization has nonvanishing factor g near z_0, so the only nearby zero is the explicit factor z_0. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 7.18 (Identity theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two holomorphic functions on a connected domain agree on a set with an accumulation point in the domain, then they agree everywhere.

Solution. Their difference is holomorphic and has a nonisolated zero. The isolated-zero theorem forces the difference to vanish identically on the connected domain. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 7.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Zeros of holomorphic functions are isolated unless the function vanishes identically. This local factorization yields the identity theorem, uniqueness from accumulation sets, and multiplicity as the correct notion of zero order. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

7.12 Exercises with complete solutions

Exercise 7.20. State and apply the central principle behind Zeros and multiplicity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. If $f(z_0) = 0$, the smallest nonzero Taylor coefficient determines the order of the zero. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.21. State and apply the central principle behind Local factorization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Near a zero of order m, write $f(z) = (z - z_0)^m g(z)$ with g holomorphic and nonzero at z_0. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.22. State and apply the central principle behind Isolated zeros. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A nonzero holomorphic function has isolated zeros. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.23. State and apply the central principle behind Identity theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Agreement on a set with an interior accumulation point forces two holomorphic functions to agree identically on a connected domain. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.24. State and apply the central principle behind Uniqueness of analytic continuation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A holomorphic continuation, when it exists on a connected overlap, is unique. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.25. State and apply the central principle behind Zero sets. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Nontrivial holomorphic zero sets in one complex variable are discrete. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.26. State and apply the central principle behind Multiplicity under products. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Orders of zeros add under multiplication and subtract under quotients when the denominator order is smaller. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 7.27. State and apply the central principle behind Applications. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Functional identities, symmetry extensions, and polynomial root counts all use the identity principle. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 8

Laurent Series

Laurent series extend power-series expansions to annuli and encode singular behavior through negative powers. The principal part distinguishes removable singularities, poles, and essential singularities.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

8.1 Laurent expansions

A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$.

8.2 Annuli of convergence

Different singularities determine concentric annuli on which different Laurent expansions may be valid.

8.3 Coefficient formula

Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$.

8.4 Principal part

Negative-power terms constitute the principal part and measure the singular component.

8.5 Positive part

Nonnegative powers form an ordinary Taylor series.

8.6 Geometric expansions

Rational functions are expanded by rewriting factors into geometric-series form suited to the chosen annulus.

8.7 Uniqueness

The Laurent coefficients are uniquely determined by contour integrals.

8.8 Residue preview

The coefficient a_{-1} is the residue and controls contour integrals around the singularity.

8.9 Annular deformation

Coefficient integrals are independent of the chosen circle within the annulus.

8.10 Core structural results

Theorem 8.1 (Laurent theorem). *A function holomorphic on an annulus $r < |z - z_0| < R$ has a unique Laurent series converging normally on compact subannuli.*

Proof. Apply Cauchy's formula to a region bounded by two circles, expand the outer and inner Cauchy kernels geometrically in the appropriate ratios, and justify termwise integration by uniform convergence on compact subannuli. □

Theorem 8.2 (Laurent coefficient formula). *For any circle $|\zeta - z_0| = \rho$ inside the annulus, $a_n = \frac{1}{2\pi i} \int \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$.*

Proof. Insert the normally convergent Laurent series into the integral and use orthogonality of powers around the circle; only the nth coefficient survives. □

Theorem 8.3 (Uniqueness). *A Laurent expansion on a fixed annulus is unique.*

Proof. The coefficient integral recovers every coefficient directly from the function, so two expansions must have identical coefficients. □

8.11 Worked examples

Example 8.4 (Laurent expansions diagnostic). Give a rigorous worked analysis of Laurent expansions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 8.5 (Annuli of convergence diagnostic). Give a rigorous worked analysis of Annuli of convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Different singularities determine concentric annuli on which different Laurent expansions may be valid. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 8.6 (Coefficient formula diagnostic)**.** Give a rigorous worked analysis of Coefficient formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 8.7 (Principal part diagnostic)**.** Give a rigorous worked analysis of Principal part. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Negative-power terms constitute the principal part and measure the singular component. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

8.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 8.8 (Laurent expansions diagnostic)**.** Give a rigorous worked analysis of Laurent expansions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.9 (Annuli of convergence diagnostic)**.** Give a rigorous worked analysis of Annuli of convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Different singularities determine concentric annuli on which different Laurent expansions may be valid. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.10 (Coefficient formula diagnostic)**.** Give a rigorous worked analysis of Coefficient formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.11 (Principal part diagnostic)**.** Give a rigorous worked analysis of Principal part. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Negative-power terms constitute the principal part and measure the singular component. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.12 (Positive part diagnostic)**.** Give a rigorous worked analysis of Positive part. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Nonnegative powers form an ordinary Taylor series. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.13 (Geometric expansions diagnostic)**.** Give a rigorous worked analysis of Geometric expansions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Rational functions are expanded by rewriting factors into geometric-series form suited to the chosen annulus. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.14 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The Laurent coefficients are uniquely determined by contour integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.15 (Residue preview diagnostic)**.** Give a rigorous worked analysis of Residue preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The coefficient a_{-1} is the residue and controls contour integrals around the singularity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 8.16 (Laurent theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A function holomorphic on an annulus $r < |z - z_0| < R$ has a unique Laurent series converging normally on compact subannuli.

Solution. Apply Cauchy's formula to a region bounded by two circles, expand the outer and inner Cauchy kernels geometrically in the appropriate ratios, and justify termwise integration by uniform convergence on compact subannuli. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 8.17 (Laurent coefficient formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For any circle $|\zeta - z_0| = \rho$ inside the annulus, $a_n = \frac{1}{2\pi i} \int \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$.

Solution. Insert the normally convergent Laurent series into the integral and use orthogonality of powers around the circle; only the nth coefficient survives. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 8.18 (Uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A Laurent expansion on a fixed annulus is unique.

Solution. The coefficient integral recovers every coefficient directly from the function, so two expansions must have identical coefficients. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 8.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Laurent series extend power-series expansions to annuli and encode singular behavior through negative powers. The principal part distinguishes removable singularities, poles, and essential singularities. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

8.13 Exercises with complete solutions

Exercise 8.20. State and apply the central principle behind Laurent expansions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A function holomorphic on an annulus has a unique expansion $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 8.21. State and apply the central principle behind Annuli of convergence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Different singularities determine concentric annuli on which different Laurent expansions may be valid. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 8.22. State and apply the central principle behind Coefficient formula. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Laurent coefficients are contour integrals $a_n = (2\pi i)^{-1} \int f(\zeta)(\zeta - z_0)^{-n-1} d\zeta$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 8.23. State and apply the central principle behind Principal part. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Negative-power terms constitute the principal part and measure the singular component. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 8.24. State and apply the central principle behind Positive part. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Nonnegative powers form an ordinary Taylor series. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 8.25. State and apply the central principle behind Geometric expansions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Rational functions are expanded by rewriting factors into geometric-series form suited to the chosen annulus. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 8.26. State and apply the central principle behind Uniqueness. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The Laurent coefficients are uniquely determined by contour integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 8.27. State and apply the central principle behind Residue preview. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The coefficient a_{-1} is the residue and controls contour integrals around the singularity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 9

Isolated Singularities

An isolated singularity is classified by its Laurent principal part. Removable singularities have no negative powers, poles have finitely many, and essential singularities have infinitely many.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

9.1 Removable singularities

A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture.

9.2 Poles

A pole of order m is characterized by $(z - z_0)^m f(z)$ extending holomorphically and nonzero at z_0.

9.3 Essential singularities

An essential singularity has infinitely many negative Laurent coefficients.

9.4 Classification theorem

Every isolated singularity is exactly one of removable, pole, or essential.

9.5 Riemann removable theorem

Boundedness near the puncture is enough for removability.

9.6 Casorati–Weierstrass

Near an essential singularity the image is dense in the complex plane.

9.7 Behavior of reciprocals

Zeros of a holomorphic function correspond to poles of its reciprocal, with matching order.

9.8 Singularity at infinity

Use $w = 1/z$ to classify behavior at infinity for functions on exterior domains.

9.9 Core structural results

Theorem 9.1 (Classification by Laurent series)**.** *An isolated singularity is removable iff all negative Laurent coefficients vanish, a pole iff only finitely many negative coefficients are nonzero, and essential otherwise.*

Proof. The three cases correspond exactly to whether the Laurent series extends as a Taylor series, has a finite principal part with highest negative order, or has an infinite principal part. □

Theorem 9.2 (Riemann removable singularity theorem)**.** *If f is holomorphic and bounded on a punctured disk around z_0, then f extends holomorphically to z_0.*

Proof. Use Cauchy's formula on a punctured annulus and estimate negative Laurent coefficients; boundedness forces every negative coefficient to vanish. □

Theorem 9.3 (Pole-zero reciprocity)**.** *f has a zero of order m at z_0 iff $1/f$ has a pole of order m there.*

Proof. Factor $f = (z - z_0)^m g$ with $g(z_0) \neq 0$ and invert to obtain $(1/f) = (z - z_0)^{-m}(1/g)$ with holomorphic nonvanishing factor. □

9.10 Worked examples

Example 9.4 (Removable singularities diagnostic)**.** Give a rigorous worked analysis of Removable singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 9.5 (Poles diagnostic)**.** Give a rigorous worked analysis of Poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A pole of order m is characterized by $(z - z_0)^m f(z)$ extending holomorphically and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 9.6 (Essential singularities diagnostic)**.** Give a rigorous worked analysis of Essential singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. An essential singularity has infinitely many negative Laurent coefficients. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 9.7 (Classification theorem diagnostic)**.** Give a rigorous worked analysis of Classification theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Every isolated singularity is exactly one of removable, pole, or essential. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

9.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 9.8 (Removable singularities diagnostic)**.** Give a rigorous worked analysis of Removable singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.9 (Poles diagnostic)**.** Give a rigorous worked analysis of Poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A pole of order m is characterized by $(z-z_0)^m f(z)$ extending holomorphically and nonzero at z_0. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.10 (Essential singularities diagnostic)**.** Give a rigorous worked analysis of Essential singularities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. An essential singularity has infinitely many negative Laurent coefficients. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.11 (Classification theorem diagnostic)**.** Give a rigorous worked analysis of Classification theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Every isolated singularity is exactly one of removable, pole, or essential. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.12 (Riemann removable theorem diagnostic)**.** Give a rigorous worked analysis of Riemann removable theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Boundedness near the puncture is enough for removability. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.13 (Casorati–Weierstrass diagnostic). Give a rigorous worked analysis of Casorati–Weierstrass. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near an essential singularity the image is dense in the complex plane. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.14 (Behavior of reciprocals diagnostic). Give a rigorous worked analysis of Behavior of reciprocals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Zeros of a holomorphic function correspond to poles of its reciprocal, with matching order. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.15 (Singularity at infinity diagnostic). Give a rigorous worked analysis of Singularity at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Use $w = 1/z$ to classify behavior at infinity for functions on exterior domains. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 9.16 (Classification by Laurent series proof dossier). Prove the following theorem-level statement and identify the essential hypothesis: An isolated singularity is removable iff all negative Laurent coefficients vanish, a pole iff only finitely many negative coefficients are nonzero, and essential otherwise.

Solution. The three cases correspond exactly to whether the Laurent series extends as a Taylor series, has a finite principal part with highest negative order, or has an infinite principal part. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 9.17 (Riemann removable singularity theorem proof dossier). Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic and bounded on a punctured disk around z_0, then f extends holomorphically to z_0.

Solution. Use Cauchy's formula on a punctured annulus and estimate negative Laurent coefficients; boundedness forces every negative coefficient to vanish. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 9.18 (Pole-zero reciprocity proof dossier). Prove the following theorem-level statement and identify the essential hypothesis: f has a zero of order m at z_0 iff $1/f$ has a pole of order m there.

Solution. Factor $f = (z - z_0)^m g$ with $g(z_0) \neq 0$ and invert to obtain $(1/f) = (z - z_0)^{-m}(1/g)$ with holomorphic nonvanishing factor. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 9.19 (Chapter synthesis). Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. An isolated singularity is classified by its Laurent principal part. Removable singularities have no negative powers, poles have finitely many, and essential singularities have infinitely many. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

9.12 Exercises with complete solutions

Exercise 9.20. State and apply the central principle behind Removable singularities. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A bounded holomorphic function on a punctured neighborhood extends holomorphically across the puncture. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.21. State and apply the central principle behind Poles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A pole of order m is characterized by $(z-z_0)^m f(z)$ extending holomorphically and nonzero at z_0. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.22. State and apply the central principle behind Essential singularities. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. An essential singularity has infinitely many negative Laurent coefficients. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.23. State and apply the central principle behind Classification theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Every isolated singularity is exactly one of removable, pole, or essential. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.24. State and apply the central principle behind Riemann removable theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Boundedness near the puncture is enough for removability. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.25. State and apply the central principle behind Casorati–Weierstrass. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Near an essential singularity the image is dense in the complex plane. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.26. State and apply the central principle behind Behavior of reciprocals. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Zeros of a holomorphic function correspond to poles of its reciprocal, with matching order. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 9.27. State and apply the central principle behind Singularity at infinity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Use $w = 1/z$ to classify behavior at infinity for functions on exterior domains. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 10

Residues and the Residue Theorem

Residues compress the local singular data relevant to contour integration into one Laurent coefficient. The residue theorem converts global contour integrals into finite sums of local contributions.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

10.1 Residue definition

The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$.

10.2 Simple poles

For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$.

10.3 Higher-order poles

Derivative formulas compute residues at poles of finite order.

10.4 Residue theorem

A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form.

10.5 Local-to-global principle

Only the -1 Laurent terms contribute to closed contour integrals.

10.6 Residue at infinity

For meromorphic functions on the sphere, the residue at infinity balances the finite residues.

10.7 Logarithmic derivative

Residues of f'/f record zero and pole multiplicities.

10.8 Computational strategy

Factor denominators, classify enclosed poles, choose the simplest residue formula, and audit contour orientation.

10.9 Parameter dependence

Residue calculations often reduce integral identities to algebraic sums depending analytically on parameters.

10.10 Core structural results

Theorem 10.1 (Residue theorem)**.** *If f is holomorphic inside and on a positively oriented contour except for finitely many isolated singularities a_k, then $\int_\Gamma f\,dz = 2\pi i \sum_k \operatorname{Res}(f, a_k)$ in the simple winding-one setting.*

Proof. Remove small disks around the singularities, apply Cauchy's theorem to the remaining multiply connected region, and let the small circles shrink; each circle integral tends to $2\pi i$ times the Laurent -1 coefficient. □

Theorem 10.2 (Simple-pole formula)**.** *If $f = g/h$ with g, h holomorphic, $h(a) = 0$, and $h'(a) \neq 0$, then $\operatorname{Res}(f, a) = g(a)/h'(a)$.*

Proof. Factor $h(z) = (z-a)k(z)$ with $k(a) = h'(a)$; then $(z-a)f(z) = g(z)/k(z)$ and evaluate at a. □

Theorem 10.3 (Higher-pole residue formula)**.** *If f has a pole of order m at a, then $\operatorname{Res}(f, a) = \frac{1}{(m-1)!}\left[\frac{d^{m-1}}{dz^{m-1}}((z-a)^m f(z))\right]_{z=a}$.*

Proof. The bracketed factor is holomorphic; its Taylor coefficient of degree $m-1$ is exactly the Laurent coefficient multiplying $(z-a)^{-1}$. □

Theorem 10.4 (Global residue balance)**.** *For a meromorphic function on the Riemann sphere, the sum of all residues including infinity is zero.*

Proof. Integrate around a sufficiently large circle and compare the exterior coordinate $w = 1/z$ with the finite residues inside; the residue at infinity is defined to cancel the large-circle contribution. □

10.11 Worked examples

Example 10.5 (Residue definition diagnostic)**.** Give a rigorous worked analysis of Residue definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 10.6 (Simple poles diagnostic)**.** Give a rigorous worked analysis of Simple poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 10.7 (Higher-order poles diagnostic)**.** Give a rigorous worked analysis of Higher-order poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Derivative formulas compute residues at poles of finite order. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 10.8 (Residue theorem diagnostic)**.** Give a rigorous worked analysis of Residue theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

10.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 10.9 (Residue definition diagnostic)**.** Give a rigorous worked analysis of Residue definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.10 (Simple poles diagnostic)**.** Give a rigorous worked analysis of Simple poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.11 (Higher-order poles diagnostic)**.** Give a rigorous worked analysis of Higher-order poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Derivative formulas compute residues at poles of finite order. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.12 (Residue theorem diagnostic)**.** Give a rigorous worked analysis of Residue theorem. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.13 (Local-to-global principle diagnostic)**.** Give a rigorous worked analysis of Local-to-global principle. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Only the -1 Laurent terms contribute to closed contour integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.14 (Residue at infinity diagnostic)**.** Give a rigorous worked analysis of Residue at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For meromorphic functions on the sphere, the residue at infinity balances the finite residues. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.15 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Residues of f'/f record zero and pole multiplicities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 10.16 (Residue theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic inside and on a positively oriented contour except for finitely many isolated singularities a_k, then $\int_\Gamma f dz = 2\pi i \sum_k \operatorname{Res}(f, a_k)$ in the simple winding-one setting.

Solution. Remove small disks around the singularities, apply Cauchy's theorem to the remaining multiply connected region, and let the small circles shrink; each circle integral tends to $2\pi i$ times the Laurent -1 coefficient. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 10.17 (Simple-pole formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f = g/h$ with g, h holomorphic, $h(a) = 0$, and $h'(a) \neq 0$, then $\operatorname{Res}(f, a) = g(a)/h'(a)$.

Solution. Factor $h(z) = (z - a)k(z)$ with $k(a) = h'(a)$; then $(z - a)f(z) = g(z)/k(z)$ and evaluate at a. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 10.18 (Higher-pole residue formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f has a pole of order m at a, then $\operatorname{Res}(f, a) = \frac{1}{(m-1)!} \left[\frac{d^{m-1}}{dz^{m-1}}((z - a)^m f(z))\right]_{z=a}$.

Solution. The bracketed factor is holomorphic; its Taylor coefficient of degree $m-1$ is exactly the Laurent coefficient multiplying $(z - a)^{-1}$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 10.19 (Global residue balance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a meromorphic function on the Riemann sphere, the sum of all residues including infinity is zero.

Solution. Integrate around a sufficiently large circle and compare the exterior coordinate $w = 1/z$ with the finite residues inside; the residue at infinity is defined to cancel the large-circle contribution. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 10.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Residues compress the local singular data relevant to contour integration into one Laurent coefficient. The residue theorem converts global contour integrals into finite sums of local contributions. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

10.13 Exercises with complete solutions

Exercise 10.21. State and apply the central principle behind Residue definition. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The residue at an isolated singularity is the Laurent coefficient of $(z - z_0)^{-1}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 10.22. State and apply the central principle behind Simple poles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For a simple pole of g/h with $h(z_0) = 0$ and $h'(z_0) \neq 0$, the residue is $g(z_0)/h'(z_0)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 10.23. State and apply the central principle behind Higher-order poles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Derivative formulas compute residues at poles of finite order. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 10.24. State and apply the central principle behind Residue theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A contour integral equals $2\pi i$ times the sum of enclosed residues, weighted by winding number in the general form. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 10.25. State and apply the central principle behind Local-to-global principle. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Only the -1 Laurent terms contribute to closed contour integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.26. State and apply the central principle behind Residue at infinity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. For meromorphic functions on the sphere, the residue at infinity balances the finite residues. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.27. State and apply the central principle behind Logarithmic derivative. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Residues of f'/f record zero and pole multiplicities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 10.28. State and apply the central principle behind Computational strategy. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Factor denominators, classify enclosed poles, choose the simplest residue formula, and audit contour orientation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 11

Evaluation of Real Integrals

Real integrals can often be evaluated by embedding the integrand into a meromorphic complex function and choosing contours whose auxiliary pieces vanish or are explicitly computable.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

11.1 Contour selection

Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries.

11.2 Even rational integrals

Symmetry can reduce whole-line contour integrals to half-line real integrals.

11.3 Decay on large arcs

Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish.

11.4 Jordan-type estimates

Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign.

11.5 Principal values

Poles on the contour require indentation and lead naturally to Cauchy principal values.

11.6 Trigonometric integrals

The substitution $z = e^{it}$ converts periodic rational trigonometric integrals into unit-circle contour integrals.

11.7 Parameter differentiation

Once an integral depends analytically on a parameter, differentiating a known contour identity can generate related formulas.

11.8 Branch-aware previews

Integrals with noninteger powers or logarithms require keyhole contours and branch choices, developed later.

11.9 Verification

Residue methods should be checked against convergence, orientation, enclosed singularities, and real-valued symmetry.

11.10 Core structural results

Theorem 11.1 (Upper-half-plane residue method)**.** *Suppose a rational function has no real poles and decays fast enough that the upper semicircle contribution vanishes. Then its real-line integral equals $2\pi i$ times the residues of poles in the upper half-plane.*

Proof. Apply the residue theorem to the closed contour consisting of the real segment and upper semicircle, then send the radius to infinity using the decay estimate. □

Theorem 11.2 (Unit-circle substitution)**.** *For a 2π-periodic rational expression in $\sin t$ and $\cos t$, the substitution $z = e^{it}$ converts the integral into a contour integral on $|z| = 1$.*

Proof. Use $dt = dz/(iz)$, $\cos t = (z + z^{-1})/2$, and $\sin t = (z - z^{-1})/(2i)$; simplify to a rational function of z and apply residues inside the unit circle. □

Theorem 11.3 (Principal-value indentation rule)**.** *A simple pole on a smooth contour contributes half of the usual residue term under a symmetric indentation, with sign determined by the indentation orientation.*

Proof. Replace a small neighborhood of the pole by a semicircular detour, approximate the integrand by its residue divided by $z - a$, and compute the limiting half-circle integral. □

11.11 Worked examples

Example 11.4 (Contour selection diagnostic)**.** Give a rigorous worked analysis of Contour selection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 11.5 (Even rational integrals diagnostic)**.** Give a rigorous worked analysis of Even rational integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Symmetry can reduce whole-line contour integrals to half-line real integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 11.6 (Decay on large arcs diagnostic)**.** Give a rigorous worked analysis of Decay on large arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 11.7 (Jordan-type estimates diagnostic)**.** Give a rigorous worked analysis of Jordan-type estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

11.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 11.8 (Contour selection diagnostic)**.** Give a rigorous worked analysis of Contour selection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.9 (Even rational integrals diagnostic)**.** Give a rigorous worked analysis of Even rational integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Symmetry can reduce whole-line contour integrals to half-line real integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.10 (Decay on large arcs diagnostic)**.** Give a rigorous worked analysis of Decay on large arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.11 (Jordan-type estimates diagnostic)**.** Give a rigorous worked analysis of Jordan-type estimates. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.12 (Principal values diagnostic)**.** Give a rigorous worked analysis of Principal values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Poles on the contour require indentation and lead naturally to Cauchy principal values. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.13 (Trigonometric integrals diagnostic)**.** Give a rigorous worked analysis of Trigonometric integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The substitution $z = e^{it}$ converts periodic rational trigonometric integrals into unit-circle contour integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.14 (Parameter differentiation diagnostic)**.** Give a rigorous worked analysis of Parameter differentiation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Once an integral depends analytically on a parameter, differentiating a known contour identity can generate related formulas. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.15 (Branch-aware previews diagnostic)**.** Give a rigorous worked analysis of Branch-aware previews. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Integrals with noninteger powers or logarithms require keyhole contours and branch choices, developed later. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 11.16 (Upper-half-plane residue method proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Suppose a rational function has no real poles and decays fast enough that the upper semicircle contribution vanishes. Then its real-line integral equals $2\pi i$ times the residues of poles in the upper half-plane.

Solution. Apply the residue theorem to the closed contour consisting of the real segment and upper semicircle, then send the radius to infinity using the decay estimate. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 11.17 (Unit-circle substitution proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a 2π-periodic rational expression in $\sin t$ and $\cos t$, the substitution $z = e^{it}$ converts the integral into a contour integral on $|z| = 1$.

Solution. Use $dt = dz/(iz)$, $\cos t = (z + z^{-1})/2$, and $\sin t = (z - z^{-1})/(2i)$; simplify to a rational function of z and apply residues inside the unit circle. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 11.18 (Principal-value indentation rule proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A simple pole on a smooth contour contributes half of the usual residue term under a symmetric indentation, with sign determined by the indentation orientation.

Solution. Replace a small neighborhood of the pole by a semicircular detour, approximate the integrand by its residue divided by $z - a$, and compute the limiting half-circle integral. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 11.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Real integrals can often be evaluated by embedding the integrand into a meromorphic complex function and choosing contours whose auxiliary pieces vanish or are explicitly computable. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

11.13 Exercises with complete solutions

Exercise 11.20. State and apply the central principle behind Contour selection. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Semicircles suit rational decay on the real line; sectors and rectangles exploit exponential or trigonometric symmetries. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.21. State and apply the central principle behind Even rational integrals. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Symmetry can reduce whole-line contour integrals to half-line real integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.22. State and apply the central principle behind Decay on large arcs. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Degree comparison or exponential estimates determine whether auxiliary arc integrals vanish. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.23. State and apply the central principle behind Jordan-type estimates. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Exponential factors e^{iaz} decay on suitable half-planes when a has the correct sign. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.24. State and apply the central principle behind Principal values. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Poles on the contour require indentation and lead naturally to Cauchy principal values. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.25. State and apply the central principle behind Trigonometric integrals. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The substitution $z = e^{it}$ converts periodic rational trigonometric integrals into unit-circle contour integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.26. State and apply the central principle behind Parameter differentiation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Once an integral depends analytically on a parameter, differentiating a known contour identity can generate related formulas. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 11.27. State and apply the central principle behind Branch-aware previews. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Integrals with noninteger powers or logarithms require keyhole contours and branch choices, developed later. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Part III

Global Complex Analysis

Chapter 12

Winding Numbers and the Argument Principle

Winding numbers measure how closed curves wrap around points, while the argument principle converts winding into an exact count of zeros and poles. This chapter is the bridge from contour integration to global counting theorems.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

12.1 Index of a closed curve

For a closed curve Γ avoiding a, the index $\operatorname{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a.

12.2 Homotopy invariance

The winding number is unchanged under deformations of the curve that avoid the reference point.

12.3 Local constancy

As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement.

12.4 Logarithmic derivative

For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities.

12.5 Argument principle

The contour integral of f'/f counts zeros minus poles inside a contour, weighted by multiplicity.

12.6 Change of argument

The argument principle can be interpreted as total variation of the argument of $f(\Gamma)$.

12.7 Counting polynomial roots

Contours can count roots in regions without locating them explicitly.

12.8 General residue formulation

The winding-number form of the residue theorem allows multiply wound contours and non-simple geometry.

12.9 Boundary hypotheses

The argument principle requires the contour to avoid zeros and poles of the meromorphic function.

12.10 Topological synthesis

Winding, residues, multiplicity, and image curves are different faces of the same integer-valued invariant.

12.11 Core structural results

Theorem 12.1 (Integer-valued index)**.** *For a closed piecewise C^1 curve Γ and $a \notin \Gamma$, $\mathrm{Ind}(\Gamma, a)$ is an integer.*

Proof. Locally choose a branch of the logarithm along successive short pieces of the curve. The integral of $(z-a)^{-1}$ is the net change of logarithm, and after one circuit the logarithm can change only by an integral multiple of $2\pi i$. ☐

Theorem 12.2 (Argument principle)**.** *If f is meromorphic inside and on Γ, with no zeros or poles on Γ, then $\frac{1}{2\pi i}\int_\Gamma f'(z)/f(z)\,dz = N - P$, counting zeros and poles with multiplicity in the simple winding-one case.*

Proof. The logarithmic derivative has a simple pole at every zero with residue equal to its multiplicity and at every pole with residue minus its order. Apply the residue theorem. ☐

Theorem 12.3 (Homotopy invariance of index)**.** *If two closed curves are homotopic in $\mathbb{C} \setminus \{a\}$, then they have the same winding number about a.*

Proof. Apply homotopy invariance of contour integrals to the holomorphic function $(z-a)^{-1}$ on the punctured plane. ☐

Theorem 12.4 (Local constancy of index)**.** *The map $a \mapsto \mathrm{Ind}(\Gamma, a)$ is constant on every connected component of $\mathbb{C} \setminus \Gamma$.*

Proof. The integral depends continuously on a off the curve, while its values are integers. A continuous integer-valued function is locally constant, hence constant on connected components. $\square$

12.12 Worked examples

Example 12.5 (Index of a closed curve diagnostic)**.** Give a rigorous worked analysis of Index of a closed curve. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a closed curve Γ avoiding a, the index $\mathrm{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 12.6 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The winding number is unchanged under deformations of the curve that avoid the reference point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 12.7 (Local constancy diagnostic)**.** Give a rigorous worked analysis of Local constancy. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 12.8 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

12.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 12.9 (Index of a closed curve diagnostic)**.** Give a rigorous worked analysis of Index of a closed curve. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a closed curve Γ avoiding a, the index $\mathrm{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. $\square$

Problem 12.10 (Homotopy invariance diagnostic)**.** Give a rigorous worked analysis of Homotopy invariance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The winding number is unchanged under deformations of the curve that avoid the reference point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. $\square$

Problem 12.11 (Local constancy diagnostic)**.** Give a rigorous worked analysis of Local constancy. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.12 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.13 (Argument principle diagnostic)**.** Give a rigorous worked analysis of Argument principle. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The contour integral of f'/f counts zeros minus poles inside a contour, weighted by multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.14 (Change of argument diagnostic)**.** Give a rigorous worked analysis of Change of argument. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The argument principle can be interpreted as total variation of the argument of $f(\Gamma)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.15 (Counting polynomial roots diagnostic)**.** Give a rigorous worked analysis of Counting polynomial roots. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Contours can count roots in regions without locating them explicitly. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 12.16 (Integer-valued index proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a closed piecewise C^1 curve Γ and $a \notin \Gamma$, $\mathrm{Ind}(\Gamma, a)$ is an integer.

Solution. Locally choose a branch of the logarithm along successive short pieces of the curve. The integral of $(z-a)^{-1}$ is the net change of logarithm, and after one circuit the logarithm can change only by an integral multiple of $2\pi i$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 12.17 (Argument principle proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is meromorphic inside and on Γ, with no zeros or poles on Γ, then $\frac{1}{2\pi i}\int_\Gamma f'(z)/f(z)\,dz = N - P$, counting zeros and poles with multiplicity in the simple winding-one case.

Solution. The logarithmic derivative has a simple pole at every zero with residue equal to its multiplicity and at every pole with residue minus its order. Apply the residue theorem. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 12.18 (Homotopy invariance of index proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two closed curves are homotopic in $\mathbb{C} \setminus \{a\}$, then they have the same winding number about a.

Solution. Apply homotopy invariance of contour integrals to the holomorphic function $(z-a)^{-1}$ on the punctured plane. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 12.19 (Local constancy of index proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The map $a \mapsto \mathrm{Ind}(\Gamma, a)$ is constant on every connected component of $\mathbb{C} \setminus \Gamma$.

Solution. The integral depends continuously on a off the curve, while its values are integers. A continuous integer-valued function is locally constant, hence constant on connected components. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 12.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Winding numbers measure how closed curves wrap around points, while the argument principle converts winding into an exact count of zeros and poles. This chapter is the bridge from contour integration to global counting theorems. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

12.14 Exercises with complete solutions

Exercise 12.21. State and apply the central principle behind Index of a closed curve. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. For a closed curve Γ avoiding a, the index $\mathrm{Ind}(\Gamma, a) = \frac{1}{2\pi i}\int_\Gamma \frac{dz}{z-a}$ records the net winding around a. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.22. State and apply the central principle behind Homotopy invariance. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The winding number is unchanged under deformations of the curve that avoid the reference point. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.23. State and apply the central principle behind Local constancy. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. As a function of a off the curve, the winding number is integer-valued and locally constant on each component of the complement. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.24. State and apply the central principle behind Logarithmic derivative. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. For meromorphic f, the quotient f'/f has residues equal to zero multiplicities and negatives of pole multiplicities. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.25. State and apply the central principle behind Argument principle. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The contour integral of f'/f counts zeros minus poles inside a contour, weighted by multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.26. State and apply the central principle behind Change of argument. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The argument principle can be interpreted as total variation of the argument of $f(\Gamma)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.27. State and apply the central principle behind Counting polynomial roots. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Contours can count roots in regions without locating them explicitly. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 12.28. State and apply the central principle behind General residue formulation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The winding-number form of the residue theorem allows multiply wound contours and non-simple geometry. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 13

Rouché's Theorem

Rouché's theorem compares two holomorphic functions on a boundary and concludes that they have the same number of interior zeros. It is one of the most efficient tools for root counting and perturbative existence.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

13.1 Boundary domination

The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary.

13.2 Homotopy viewpoint

The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change.

13.3 Zero counting

The argument principle turns that invariant winding number into equality of zero counts.

13.4 Polynomial applications

Dominant monomials on circles can count roots inside disks.

13.5 Perturbation stability

Small holomorphic perturbations preserve the number of zeros inside a protected contour.

13.6 Localization

Choosing several contours can isolate different root clusters.

13.7 Comparison choices

The art is to choose the part of the function that dominates on the chosen boundary.

13.8 Multiplicity

Rouché counts zeros with multiplicity, exactly as the argument principle does.

13.9 Core structural results

Theorem 13.1 (Rouché theorem)**.** *If f, g are holomorphic inside and on a simple closed contour Γ and $|g| < |f|$ on Γ, then f and $f + g$ have the same number of zeros inside, counted with multiplicity.*

Proof. For $0 \leq t \leq 1$, $f + tg$ is nonzero on Γ because $|tg| < |f|$. Hence the winding number of $(f + tg)(\Gamma)$ about zero is constant in t. Apply the argument principle at $t = 0$ and $t = 1$. □

Theorem 13.2 (Polynomial root localization)**.** *If on $|z| = R$ one term of a polynomial strictly dominates the sum of all others, then the polynomial has the same number of zeros in $|z| < R$ as that dominant term.*

Proof. Apply Rouché with the dominant term as f and the sum of the remaining terms as g. □

Theorem 13.3 (Perturbative persistence)**.** *If f has no zeros on Γ and $\|g\|_\Gamma < \min_\Gamma |f|$, then $f + g$ has the same interior zero count as f.*

Proof. The displayed inequality is precisely the strict Rouché hypothesis. □

13.10 Worked examples

Example 13.4 (Boundary domination diagnostic)**.** Give a rigorous worked analysis of Boundary domination. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 13.5 (Homotopy viewpoint diagnostic)**.** Give a rigorous worked analysis of Homotopy viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 13.6 (Zero counting diagnostic)**.** Give a rigorous worked analysis of Zero counting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The argument principle turns that invariant winding number into equality of zero counts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 13.7 (Polynomial applications diagnostic)**.** Give a rigorous worked analysis of Polynomial applications. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Dominant monomials on circles can count roots inside disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

13.11 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 13.8 (Boundary domination diagnostic)**.** Give a rigorous worked analysis of Boundary domination. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The hypothesis $|g| < |f|$ on a contour ensures that f and $f + g$ never cross zero along the boundary. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 13.9 (Homotopy viewpoint diagnostic)**.** Give a rigorous worked analysis of Homotopy viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 13.10 (Zero counting diagnostic)**.** Give a rigorous worked analysis of Zero counting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The argument principle turns that invariant winding number into equality of zero counts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 13.11 (Polynomial applications diagnostic)**.** Give a rigorous worked analysis of Polynomial applications. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Dominant monomials on circles can count roots inside disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 13.12 (Perturbation stability diagnostic)**.** Give a rigorous worked analysis of Perturbation stability. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Small holomorphic perturbations preserve the number of zeros inside a protected contour. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 13.13 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Choosing several contours can isolate different root clusters. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.14 (Comparison choices diagnostic)**.** Give a rigorous worked analysis of Comparison choices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The art is to choose the part of the function that dominates on the chosen boundary. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.15 (Multiplicity diagnostic)**.** Give a rigorous worked analysis of Multiplicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Rouché counts zeros with multiplicity, exactly as the argument principle does. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 13.16 (Rouché theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f, g are holomorphic inside and on a simple closed contour Γ and $|g| < |f|$ on Γ, then f and $f + g$ have the same number of zeros inside, counted with multiplicity.

Solution. For $0 \leq t \leq 1$, $f + tg$ is nonzero on Γ because $|tg| < |f|$. Hence the winding number of $(f + tg)(\Gamma)$ about zero is constant in t. Apply the argument principle at $t = 0$ and $t = 1$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 13.17 (Polynomial root localization proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If on $|z| = R$ one term of a polynomial strictly dominates the sum of all others, then the polynomial has the same number of zeros in $|z| < R$ as that dominant term.

Solution. Apply Rouché with the dominant term as f and the sum of the remaining terms as g. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 13.18 (Perturbative persistence proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f has no zeros on Γ and $\|g\|_\Gamma < \min_\Gamma |f|$, then $f + g$ has the same interior zero count as f.

Solution. The displayed inequality is precisely the strict Rouché hypothesis. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 13.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Rouché's theorem compares two holomorphic functions on a boundary and concludes that they have the same number of interior zeros. It is one of the most efficient tools for root counting and perturbative existence. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

13.12 Exercises with complete solutions

Exercise 13.20. State and apply the central principle behind Boundary domination. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The hypothesis $|g| < |f|$ on a contour ensures that f and $f+g$ never cross zero along the boundary. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.21. State and apply the central principle behind Homotopy viewpoint. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The family $f + tg$ remains nonzero on the contour, so its boundary winding around zero cannot change. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.22. State and apply the central principle behind Zero counting. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The argument principle turns that invariant winding number into equality of zero counts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.23. State and apply the central principle behind Polynomial applications. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Dominant monomials on circles can count roots inside disks. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.24. State and apply the central principle behind Perturbation stability. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Small holomorphic perturbations preserve the number of zeros inside a protected contour. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.25. State and apply the central principle behind Localization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Choosing several contours can isolate different root clusters. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.26. State and apply the central principle behind Comparison choices. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The art is to choose the part of the function that dominates on the chosen boundary. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 13.27. State and apply the central principle behind Multiplicity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Rouché counts zeros with multiplicity, exactly as the argument principle does. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 14

Branches of the Logarithm and Roots

Branches of logarithms and roots exist exactly when the topology of the domain allows one to choose arguments continuously. The obstruction is winding around the origin.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

14.1 Multivalued logarithm

Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$.

14.2 Branches

A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$.

14.3 Argument functions

Writing $L(z) = \log|z| + i\,\mathrm{Arg}(z)$ links branches of logarithm to continuous arguments.

14.4 Slit domains

Removing a ray from the plane yields standard simply connected domains supporting logarithm branches.

14.5 Nth roots

A branch of $z^{1/n}$ is obtained from a logarithm branch by exponentiation.

14.6 Powers

For complex exponent α, define $z^\alpha = e^{\alpha L(z)}$ relative to a chosen logarithm branch.

14.7 Monodromy

Analytic continuation around the origin changes the logarithm by $2\pi i$ and roots by roots of unity.

14.8 Integral criterion

A logarithm branch exists when the integral of $1/z$ around every closed contour in the domain vanishes.

14.9 Topological obstruction

Nonzero winding around zero prevents a globally single-valued logarithm.

14.10 Core structural results

Theorem 14.1 (Logarithm branch criterion)**.** *On a domain $D \subset \mathbb{C} \setminus \{0\}$, a holomorphic logarithm exists if $\int_\gamma dz/z = 0$ for every closed contour γ in D.*

Proof. The vanishing integral condition gives a primitive L of $1/z$. Then $e^{-L(z)}z$ has derivative zero, so after adding a constant to L one obtains $e^{L(z)} = z$. ☐

Theorem 14.2 (Simply connected logarithm theorem)**.** *Every simply connected domain avoiding zero admits a holomorphic logarithm.*

Proof. The function $1/z$ is holomorphic on the domain, and Cauchy's theorem makes all its closed contour integrals vanish. Apply the logarithm branch criterion. ☐

Theorem 14.3 (Root from logarithm)**.** *If L is a logarithm branch on D, then $R(z) = e^{L(z)/n}$ is a holomorphic branch of the nth root.*

Proof. Directly, $R(z)^n = e^{L(z)} = z$, and holomorphicity follows from composition. ☐

14.11 Worked examples

Example 14.4 (Multivalued logarithm diagnostic)**.** Give a rigorous worked analysis of Multivalued logarithm. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 14.5 (Branches diagnostic)**.** Give a rigorous worked analysis of Branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 14.6 (Argument functions diagnostic)**.** Give a rigorous worked analysis of Argument functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Writing $L(z) = \log|z| + i\,\mathrm{Arg}(z)$ links branches of logarithm to continuous arguments. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 14.7 (Slit domains diagnostic)**.** Give a rigorous worked analysis of Slit domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Removing a ray from the plane yields standard simply connected domains supporting logarithm branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

14.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 14.8 (Multivalued logarithm diagnostic)**.** Give a rigorous worked analysis of Multivalued logarithm. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.9 (Branches diagnostic)**.** Give a rigorous worked analysis of Branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.10 (Argument functions diagnostic)**.** Give a rigorous worked analysis of Argument functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Writing $L(z) = \log|z|+i\,\mathrm{Arg}(z)$ links branches of logarithm to continuous arguments. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.11 (Slit domains diagnostic)**.** Give a rigorous worked analysis of Slit domains. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Removing a ray from the plane yields standard simply connected domains supporting logarithm branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 14.12 (Nth roots diagnostic)**.** Give a rigorous worked analysis of Nth roots. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A branch of $z^{1/n}$ is obtained from a logarithm branch by exponentiation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 14.13 (Powers diagnostic)**.** Give a rigorous worked analysis of Powers. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For complex exponent α, define $z^\alpha = e^{\alpha L(z)}$ relative to a chosen logarithm branch. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 14.14 (Monodromy diagnostic)**.** Give a rigorous worked analysis of Monodromy. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Analytic continuation around the origin changes the logarithm by $2\pi i$ and roots by roots of unity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 14.15 (Integral criterion diagnostic)**.** Give a rigorous worked analysis of Integral criterion. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A logarithm branch exists when the integral of $1/z$ around every closed contour in the domain vanishes. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 14.16 (Logarithm branch criterion proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: On a domain $D \subset \mathbb{C} \setminus \{0\}$, a holomorphic logarithm exists if $\int_\gamma dz/z = 0$ for every closed contour γ in D.

Solution. The vanishing integral condition gives a primitive L of $1/z$. Then $e^{-L(z)}z$ has derivative zero, so after adding a constant to L one obtains $e^{L(z)} = z$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 14.17 (Simply connected logarithm theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every simply connected domain avoiding zero admits a holomorphic logarithm.

Solution. The function $1/z$ is holomorphic on the domain, and Cauchy's theorem makes all its closed contour integrals vanish. Apply the logarithm branch criterion. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 14.18 (Root from logarithm proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If L is a logarithm branch on D, then $R(z) = e^{L(z)/n}$ is a holomorphic branch of the nth root.

Solution. Directly, $R(z)^n = e^{L(z)} = z$, and holomorphicity follows from composition. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 14.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Branches of logarithms and roots exist exactly when the topology of the domain allows one to choose arguments continuously. The obstruction is winding around the origin. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

14.13 Exercises with complete solutions

Exercise 14.20. State and apply the central principle behind Multivalued logarithm. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Solving $e^w = z$ gives values differing by integer multiples of $2\pi i$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 14.21. State and apply the central principle behind Branches. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A branch of logarithm on a domain is a holomorphic function L with $e^{L(z)} = z$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 14.22. State and apply the central principle behind Argument functions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Writing $L(z) = \log|z| + i\,\mathrm{Arg}(z)$ links branches of logarithm to continuous arguments. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 14.23. State and apply the central principle behind Slit domains. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Removing a ray from the plane yields standard simply connected domains supporting logarithm branches. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 14.24. State and apply the central principle behind Nth roots. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A branch of $z^{1/n}$ is obtained from a logarithm branch by exponentiation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 14.25. State and apply the central principle behind Powers. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. For complex exponent α, define $z^\alpha = e^{\alpha L(z)}$ relative to a chosen logarithm branch. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.26. State and apply the central principle behind Monodromy. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Analytic continuation around the origin changes the logarithm by $2\pi i$ and roots by roots of unity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 14.27. State and apply the central principle behind Integral criterion. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A logarithm branch exists when the integral of $1/z$ around every closed contour in the domain vanishes. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 15

Analytic Continuation

Analytic continuation extends local holomorphic data across overlapping domains. The identity theorem guarantees uniqueness, while singularities and monodromy determine how far continuation can proceed.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

15.1 Germs

A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods.

15.2 Direct continuation

Two local representatives continue one another when they agree on a connected overlap.

15.3 Chains of continuation

A function can be propagated through a sequence of overlapping disks.

15.4 Uniqueness

The identity theorem prevents two different continuations from branching on the same connected overlap.

15.5 Continuation along paths

Local elements can be transported step by step along a curve when singularities are avoided.

15.6 Natural boundaries

Some power series encounter singularities densely along their circle of convergence and cannot continue through it.

15.7 Singular obstruction

Poles, branch points, and essential singularities can stop single-valued continuation.

15.8 Monodromy preview

Continuation around loops may return a different branch unless topology forces uniqueness globally.

15.9 Maximal continuation

The collection of all continuations naturally suggests a Riemann surface rather than a single planar domain.

15.10 Core structural results

Theorem 15.1 (Uniqueness of analytic continuation)**.** *If two holomorphic continuations agree on one nonempty open subset of a connected overlap, then they agree on the entire overlap.*

Proof. Their difference is holomorphic and vanishes on an open set, hence by the identity theorem vanishes everywhere on the connected overlap. ☐

Theorem 15.2 (Continuation along a fixed chain is unique)**.** *Given a chain of connected overlapping domains and an initial germ, at most one compatible holomorphic element can occupy each successive domain.*

Proof. Apply uniqueness of analytic continuation at each overlap inductively. ☐

Theorem 15.3 (Pathwise local uniqueness)**.** *Two continuations of the same initial germ along sufficiently close subdivisions of the same path agree wherever both are defined.*

Proof. Refine the subdivisions to common overlapping neighborhoods and repeatedly apply the identity theorem. ☐

15.11 Worked examples

Example 15.4 (Germs diagnostic)**.** Give a rigorous worked analysis of Germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 15.5 (Direct continuation diagnostic)**.** Give a rigorous worked analysis of Direct continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Two local representatives continue one another when they agree on a connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 15.6 (Chains of continuation diagnostic)**.** Give a rigorous worked analysis of Chains of continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A function can be propagated through a sequence of overlapping disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 15.7 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The identity theorem prevents two different continuations from branching on the same connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

15.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 15.8 (Germs diagnostic)**.** Give a rigorous worked analysis of Germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.9 (Direct continuation diagnostic)**.** Give a rigorous worked analysis of Direct continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Two local representatives continue one another when they agree on a connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.10 (Chains of continuation diagnostic)**.** Give a rigorous worked analysis of Chains of continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A function can be propagated through a sequence of overlapping disks. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.11 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The identity theorem prevents two different continuations from branching on the same connected overlap. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.12 (Continuation along paths diagnostic)**.** Give a rigorous worked analysis of Continuation along paths. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Local elements can be transported step by step along a curve when singularities are avoided. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.13 (Natural boundaries diagnostic)**.** Give a rigorous worked analysis of Natural boundaries. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Some power series encounter singularities densely along their circle of convergence and cannot continue through it. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.14 (Singular obstruction diagnostic)**.** Give a rigorous worked analysis of Singular obstruction. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Poles, branch points, and essential singularities can stop single-valued continuation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.15 (Monodromy preview diagnostic)**.** Give a rigorous worked analysis of Monodromy preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Continuation around loops may return a different branch unless topology forces uniqueness globally. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 15.16 (Uniqueness of analytic continuation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two holomorphic continuations agree on one nonempty open subset of a connected overlap, then they agree on the entire overlap.

Solution. Their difference is holomorphic and vanishes on an open set, hence by the identity theorem vanishes everywhere on the connected overlap. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 15.17 (Continuation along a fixed chain is unique proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Given a chain of connected overlapping domains and an initial germ, at most one compatible holomorphic element can occupy each successive domain.

Solution. Apply uniqueness of analytic continuation at each overlap inductively. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 15.18 (Pathwise local uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Two continuations of the same initial germ along sufficiently close subdivisions of the same path agree wherever both are defined.

Solution. Refine the subdivisions to common overlapping neighborhoods and repeatedly apply the identity theorem. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 15.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Analytic continuation extends local holomorphic data across overlapping domains. The identity theorem guarantees uniqueness, while singularities and monodromy determine how far continuation can proceed. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

15.13 Exercises with complete solutions

Exercise 15.20. State and apply the central principle behind Germs. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A germ records the local behavior of a holomorphic function near one point, modulo agreement on smaller neighborhoods. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.21. State and apply the central principle behind Direct continuation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Two local representatives continue one another when they agree on a connected overlap. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.22. State and apply the central principle behind Chains of continuation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A function can be propagated through a sequence of overlapping disks. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.23. State and apply the central principle behind Uniqueness. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The identity theorem prevents two different continuations from branching on the same connected overlap. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.24. State and apply the central principle behind Continuation along paths. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Local elements can be transported step by step along a curve when singularities are avoided. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.25. State and apply the central principle behind Natural boundaries. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Some power series encounter singularities densely along their circle of convergence and cannot continue through it. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.26. State and apply the central principle behind Singular obstruction. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Poles, branch points, and essential singularities can stop single-valued continuation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 15.27. State and apply the central principle behind Monodromy preview. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Continuation around loops may return a different branch unless topology forces uniqueness globally. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 16

Möbius Transformations

Möbius transformations are the automorphisms of the Riemann sphere generated by fractional linear maps. They preserve generalized circles and provide the elementary coordinate changes of global complex analysis.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

16.1 Fractional linear maps

A Möbius map has form $T(z) = (az + b)/(cz + d)$ with $ad - bc \neq 0$, extended naturally to infinity.

16.2 Matrix representation

Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation.

16.3 Composition

Composition corresponds to matrix multiplication modulo nonzero scalars.

16.4 Three-point normalization

A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple.

16.5 Cross-ratio

The cross-ratio is invariant under Möbius transformations.

16.6 Generalized circles

Lines and circles on the plane are circles on the sphere, and Möbius transformations map generalized circles to generalized circles.

16.7 Elementary generators

Translations, dilations/rotations, and inversion generate all Möbius transformations.

16.8 Fixed points

Fixed points solve a quadratic equation and organize the elementary dynamics of the transformation.

16.9 Disk and half-plane maps

Specific Möbius maps give conformal equivalences between the unit disk and half-plane.

16.10 Core structural results

Theorem 16.1 (Three-point theorem)**.** *Given distinct z_1, z_2, z_3 and distinct w_1, w_2, w_3 on the Riemann sphere, there is a unique Möbius transformation with $T(z_j) = w_j$.*

Proof. Construct maps sending each triple to $(0, 1, \infty)$ using cross-ratio formulas, then compose one with the inverse of the other. Uniqueness follows because a Möbius map fixing three distinct points is the identity. □

Theorem 16.2 (Cross-ratio invariance)**.** *Möbius transformations preserve the cross-ratio of four points.*

Proof. It suffices to check translations, nonzero scalings, and inversion, which generate the Möbius group; direct algebra verifies invariance for each generator. □

Theorem 16.3 (Generalized-circle preservation)**.** *Every Möbius transformation maps circles and lines to circles or lines.*

Proof. Translations and similarities preserve both classes; inversion maps circles/lines to generalized circles by an explicit equation calculation. Use the generator decomposition. □

16.11 Worked examples

Example 16.4 (Fractional linear maps diagnostic)**.** Give a rigorous worked analysis of Fractional linear maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A Möbius map has form $T(z) = (az+b)/(cz+d)$ with $ad-bc \neq 0$, extended naturally to infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 16.5 (Matrix representation diagnostic)**.** Give a rigorous worked analysis of Matrix representation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 16.6 (Composition diagnostic)**.** Give a rigorous worked analysis of Composition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Composition corresponds to matrix multiplication modulo nonzero scalars. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 16.7 (Three-point normalization diagnostic)**.** Give a rigorous worked analysis of Three-point normalization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

16.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 16.8 (Fractional linear maps diagnostic)**.** Give a rigorous worked analysis of Fractional linear maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A Möbius map has form $T(z) = (az + b)/(cz + d)$ with $ad - bc \neq 0$, extended naturally to infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.9 (Matrix representation diagnostic)**.** Give a rigorous worked analysis of Matrix representation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.10 (Composition diagnostic)**.** Give a rigorous worked analysis of Composition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Composition corresponds to matrix multiplication modulo nonzero scalars. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.11 (Three-point normalization diagnostic)**.** Give a rigorous worked analysis of Three-point normalization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.12 (Cross-ratio diagnostic)**.** Give a rigorous worked analysis of Cross-ratio. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The cross-ratio is invariant under Möbius transformations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.13 (Generalized circles diagnostic)**.** Give a rigorous worked analysis of Generalized circles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Lines and circles on the plane are circles on the sphere, and Möbius transformations map generalized circles to generalized circles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.14 (Elementary generators diagnostic)**.** Give a rigorous worked analysis of Elementary generators. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Translations, dilations/rotations, and inversion generate all Möbius transformations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.15 (Fixed points diagnostic)**.** Give a rigorous worked analysis of Fixed points. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Fixed points solve a quadratic equation and organize the elementary dynamics of the transformation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 16.16 (Three-point theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Given distinct z_1, z_2, z_3 and distinct w_1, w_2, w_3 on the Riemann sphere, there is a unique Möbius transformation with $T(z_j) = w_j$.

Solution. Construct maps sending each triple to $(0, 1, \infty)$ using cross-ratio formulas, then compose one with the inverse of the other. Uniqueness follows because a Möbius map fixing three distinct points is the identity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 16.17 (Cross-ratio invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Möbius transformations preserve the cross-ratio of four points.

Solution. It suffices to check translations, nonzero scalings, and inversion, which generate the Möbius group; direct algebra verifies invariance for each generator. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 16.18 (Generalized-circle preservation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every Möbius transformation maps circles and lines to circles or lines.

Solution. Translations and similarities preserve both classes; inversion maps circles/lines to generalized circles by an explicit equation calculation. Use the generator decomposition. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 16.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Möbius transformations are the automorphisms of the Riemann sphere generated by fractional linear maps. They preserve generalized circles and provide the elementary coordinate changes of global complex analysis. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

16.13 Exercises with complete solutions

Exercise 16.20. State and apply the central principle behind Fractional linear maps. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A Möbius map has form $T(z) = (az+b)/(cz+d)$ with $ad-bc \neq 0$, extended naturally to infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.21. State and apply the central principle behind Matrix representation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Scalar multiples of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ define the same Möbius transformation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.22. State and apply the central principle behind Composition. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Composition corresponds to matrix multiplication modulo nonzero scalars. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.23. State and apply the central principle behind Three-point normalization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A unique Möbius transformation sends any ordered triple of distinct sphere points to any other such triple. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.24. State and apply the central principle behind Cross-ratio. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The cross-ratio is invariant under Möbius transformations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.25. State and apply the central principle behind Generalized circles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Lines and circles on the plane are circles on the sphere, and Möbius transformations map generalized circles to generalized circles. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.26. State and apply the central principle behind Elementary generators. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Translations, dilations/rotations, and inversion generate all Möbius transformations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 16.27. State and apply the central principle behind Fixed points. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Fixed points solve a quadratic equation and organize the elementary dynamics of the transformation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 17

Conformal Mapping

Conformal maps preserve angles locally and reorganize domains without destroying holomorphic structure. Holomorphic maps with nonzero derivative are conformal, and the Riemann mapping theorem gives a universal model for simply connected proper planar domains.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

17.1 Local conformality

A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation.

17.2 Angle preservation

The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative.

17.3 Local inverse

A holomorphic function with nonzero derivative has a local holomorphic inverse.

17.4 Conformal equivalence

A biholomorphic map gives an analytic equivalence between domains.

17.5 Disk automorphisms

Automorphisms of the unit disk are Möbius transformations preserving the disk.

17.6 Half-plane equivalence

The Cayley transform identifies the upper half-plane with the unit disk.

17.7 Schwarz lemma

Holomorphic self-maps of the disk fixing zero are contractive, with a rigid equality case.

17.8 Riemann mapping theorem

Every simply connected proper plane domain is biholomorphic to the unit disk.

17.9 Normalization

Fixing one point and the argument of the derivative gives uniqueness of the Riemann map.

17.10 Boundary caution

Interior conformal equivalence does not automatically provide regular extension to arbitrary boundaries.

17.11 Core structural results

Theorem 17.1 (Holomorphic maps are conformal off critical points)**.** *If f is holomorphic and $f'(z_0) \neq 0$, then f preserves oriented angles at z_0.*

Proof. The first-order expansion is $f(z_0 + h) - f(z_0) = f'(z_0)h + o(|h|)$. Multiplication by the nonzero complex number $f'(z_0)$ rotates and scales every tangent direction by the same amount. ☐

Theorem 17.2 (Schwarz lemma)**.** *If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic and $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality at a nonzero point or in derivative only for rotations.*

Proof. Apply the maximum principle to $g(z) = f(z)/z$, with removable value $g(0) = f'(0)$. For every $r < 1$, maximum modulus on $|z| = r$ gives $|g(z)| \leq 1/r$ inside; let $r \to 1$. Equality invokes the maximum-modulus rigidity. ☐

Theorem 17.3 (Automorphisms of the disk)**.** *Every automorphism of $\mathbb{D}$ has form $e^{i\theta}(z - a)/(1 - \bar{a}z)$ for some $a \in \mathbb{D}$.*

Proof. Move the image of zero back to zero with a standard disk Möbius map. Schwarz lemma applied to the resulting automorphism and its inverse forces it to be a rotation. ☐

Theorem 17.4 (Normalized Riemann-map uniqueness)**.** *If two biholomorphisms $f, g : D \to \mathbb{D}$ send z_0 to zero and have positive real derivative at z_0, then $f = g$.*

Proof. The disk automorphism $g \circ f^{-1}$ fixes zero. By the disk automorphism classification it is a rotation, and the derivative normalization forces that rotation to be the identity. ☐

17.12 Worked examples

Example 17.5 (Local conformality diagnostic)**.** Give a rigorous worked analysis of Local conformality. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 17.6 (Angle preservation diagnostic)**.** Give a rigorous worked analysis of Angle preservation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 17.7 (Local inverse diagnostic)**.** Give a rigorous worked analysis of Local inverse. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A holomorphic function with nonzero derivative has a local holomorphic inverse. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 17.8 (Conformal equivalence diagnostic)**.** Give a rigorous worked analysis of Conformal equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A biholomorphic map gives an analytic equivalence between domains. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

17.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 17.9 (Local conformality diagnostic)**.** Give a rigorous worked analysis of Local conformality. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 17.10 (Angle preservation diagnostic)**.** Give a rigorous worked analysis of Angle preservation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 17.11 (Local inverse diagnostic)**.** Give a rigorous worked analysis of Local inverse. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A holomorphic function with nonzero derivative has a local holomorphic inverse. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 17.12 (Conformal equivalence diagnostic)**.** Give a rigorous worked analysis of Conformal equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A biholomorphic map gives an analytic equivalence between domains. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.13 (Disk automorphisms diagnostic)**.** Give a rigorous worked analysis of Disk automorphisms. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Automorphisms of the unit disk are Möbius transformations preserving the disk. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.14 (Half-plane equivalence diagnostic)**.** Give a rigorous worked analysis of Half-plane equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The Cayley transform identifies the upper half-plane with the unit disk. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.15 (Schwarz lemma diagnostic)**.** Give a rigorous worked analysis of Schwarz lemma. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphic self-maps of the disk fixing zero are contractive, with a rigid equality case. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 17.16 (Holomorphic maps are conformal off critical points proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If f is holomorphic and $f'(z_0) \neq 0$, then f preserves oriented angles at z_0.

Solution. The first-order expansion is $f(z_0 + h) - f(z_0) = f'(z_0)h + o(|h|)$. Multiplication by the nonzero complex number $f'(z_0)$ rotates and scales every tangent direction by the same amount. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 17.17 (Schwarz lemma proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic and $f(0) = 0$, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality at a nonzero point or in derivative only for rotations.

Solution. Apply the maximum principle to $g(z) = f(z)/z$, with removable value $g(0) = f'(0)$. For every $r < 1$, maximum modulus on $|z| = r$ gives $|g(z)| \leq 1/r$ inside; let $r \to 1$. Equality invokes the maximum-modulus rigidity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 17.18 (Automorphisms of the disk proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every automorphism of $\mathbb{D}$ has form $e^{i\theta}(z - a)/(1 - \bar{a}z)$ for some $a \in \mathbb{D}$.

Solution. Move the image of zero back to zero with a standard disk Möbius map. Schwarz lemma applied to the resulting automorphism and its inverse forces it to be a rotation. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 17.19 (Normalized Riemann-map uniqueness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If two biholomorphisms $f, g : D \to \mathbb{D}$ send z_0 to zero and have positive real derivative at z_0, then $f = g$.

Solution. The disk automorphism $g \circ f^{-1}$ fixes zero. By the disk automorphism classification it is a rotation, and the derivative normalization forces that rotation to be the identity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 17.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Conformal maps preserve angles locally and reorganize domains without destroying holomorphic structure. Holomorphic maps with nonzero derivative are conformal, and the Riemann mapping theorem gives a universal model for simply connected proper planar domains. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

17.14 Exercises with complete solutions

Exercise 17.21. State and apply the central principle behind Local conformality. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A holomorphic derivative $f'(z_0) \neq 0$ acts to first order as a positive dilation followed by a rotation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.22. State and apply the central principle behind Angle preservation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The argument of the quotient of tangent directions is preserved by multiplication by one nonzero complex derivative. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.23. State and apply the central principle behind Local inverse. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A holomorphic function with nonzero derivative has a local holomorphic inverse. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 17.24. State and apply the central principle behind Conformal equivalence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A biholomorphic map gives an analytic equivalence between domains. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 17.25. State and apply the central principle behind Disk automorphisms. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Automorphisms of the unit disk are Möbius transformations preserving the disk. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 17.26. State and apply the central principle behind Half-plane equivalence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The Cayley transform identifies the upper half-plane with the unit disk. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 17.27. State and apply the central principle behind Schwarz lemma. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Holomorphic self-maps of the disk fixing zero are contractive, with a rigid equality case. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 17.28. State and apply the central principle behind Riemann mapping theorem. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Every simply connected proper plane domain is biholomorphic to the unit disk. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 18

Schwarz–Christoffel Transformations

Schwarz–Christoffel transformations map canonical half-plane or disk domains to polygonal regions. The exponents in the derivative encode turning angles at polygon vertices.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

18.1 Polygonal target

A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$.

18.2 Derivative form

For a half-plane map with real prevertices x_k, the derivative has factors $(z - x_k)^{\alpha_k - 1}$.

18.3 Turning angles

The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex.

18.4 Integration

The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants.

18.5 Prevertices

Determining the prevertices is the accessory-parameter problem and is generally nonlinear.

18.6 Normalization

Three real degrees of freedom can be fixed by half-plane Möbius automorphisms.

18.7 Rectangles

Symmetry reduces rectangle maps to elliptic integrals.

18.8 Slits and degenerate polygons

Limiting polygon configurations produce maps to slit domains and strips.

18.9 Numerical construction

Practical Schwarz–Christoffel mapping often solves the parameter problem numerically.

18.10 Core structural results

Theorem 18.1 (Angle-exponent relation)**.** *Near a prevertex x_k, a Schwarz–Christoffel map with derivative behaving like $(z - x_k)^{\alpha_k - 1}$ maps a half-neighborhood to a wedge of interior angle $\alpha_k \pi$.*

Proof. Integrating gives local behavior $F(z) - F(x_k) \sim C(z - x_k)^{\alpha_k}$. The argument is therefore multiplied by α_k, sending the half-plane angle π to $\alpha_k \pi$. ☐

Theorem 18.2 (Sum of exponents)**.** *For an n-gon, $\sum_{k=1}^{n}(\alpha_k - 1) = -2$.*

Proof. The interior-angle sum is $(n-2)\pi$, so $\sum \alpha_k = n - 2$ and subtraction of n gives -2. ☐

Theorem 18.3 (Affine freedom)**.** *Once the prevertices and exponents are fixed, two Schwarz–Christoffel maps with the same derivative factors differ by a nonzero multiplicative constant and an additive constant.*

Proof. Their derivatives are proportional by the chosen normalization, so integration yields $F_2 = AF_1 + B$ with $A \neq 0$. ☐

18.11 Worked examples

Example 18.4 (Polygonal target diagnostic)**.** Give a rigorous worked analysis of Polygonal target. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A polygon is specified by ordered vertices and interior angles $\alpha_k \pi$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 18.5 (Derivative form diagnostic)**.** Give a rigorous worked analysis of Derivative form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a half-plane map with real prevertices x_k, the derivative has factors $(z - x_k)^{\alpha_k - 1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 18.6 (Turning angles diagnostic)**.** Give a rigorous worked analysis of Turning angles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 18.7 (Integration diagnostic)**.** Give a rigorous worked analysis of Integration. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

18.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 18.8 (Polygonal target diagnostic)**.** Give a rigorous worked analysis of Polygonal target. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 18.9 (Derivative form diagnostic)**.** Give a rigorous worked analysis of Derivative form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a half-plane map with real prevertices x_k, the derivative has factors $(z-x_k)^{\alpha_k-1}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 18.10 (Turning angles diagnostic)**.** Give a rigorous worked analysis of Turning angles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 18.11 (Integration diagnostic)**.** Give a rigorous worked analysis of Integration. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 18.12 (Prevertices diagnostic)**.** Give a rigorous worked analysis of Prevertices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Determining the prevertices is the accessory-parameter problem and is generally nonlinear. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.13 (Normalization diagnostic)**.** Give a rigorous worked analysis of Normalization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Three real degrees of freedom can be fixed by half-plane Möbius automorphisms. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.14 (Rectangles diagnostic)**.** Give a rigorous worked analysis of Rectangles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Symmetry reduces rectangle maps to elliptic integrals. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.15 (Slits and degenerate polygons diagnostic)**.** Give a rigorous worked analysis of Slits and degenerate polygons. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Limiting polygon configurations produce maps to slit domains and strips. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 18.16 (Angle-exponent relation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Near a prevertex x_k, a Schwarz–Christoffel map with derivative behaving like $(z - x_k)^{\alpha_k - 1}$ maps a half-neighborhood to a wedge of interior angle $\alpha_k \pi$.

Solution. Integrating gives local behavior $F(z) - F(x_k) \sim C(z - x_k)^{\alpha_k}$. The argument is therefore multiplied by α_k, sending the half-plane angle π to $\alpha_k \pi$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 18.17 (Sum of exponents proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For an n-gon, $\sum_{k=1}^{n} (\alpha_k - 1) = -2$.

Solution. The interior-angle sum is $(n-2)\pi$, so $\sum \alpha_k = n - 2$ and subtraction of n gives -2. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 18.18 (Affine freedom proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Once the prevertices and exponents are fixed, two Schwarz–Christoffel maps with the same derivative factors differ by a nonzero multiplicative constant and an additive constant.

Solution. Their derivatives are proportional by the chosen normalization, so integration yields $F_2 = AF_1 + B$ with $A \neq 0$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 18.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Schwarz–Christoffel transformations map canonical half-plane or disk domains to polygonal regions. The exponents in the derivative encode turning angles at polygon vertices. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

18.13 Exercises with complete solutions

Exercise 18.20. State and apply the central principle behind Polygonal target. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A polygon is specified by ordered vertices and interior angles $\alpha_k\pi$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.21. State and apply the central principle behind Derivative form. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For a half-plane map with real prevertices x_k, the derivative has factors $(z-x_k)^{\alpha_k-1}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.22. State and apply the central principle behind Turning angles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The exponent $\alpha_k - 1$ records the change of boundary direction at the corresponding vertex. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.23. State and apply the central principle behind Integration. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The conformal map is obtained by integrating the Schwarz–Christoffel derivative and adding affine constants. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.24. State and apply the central principle behind Prevertices. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Determining the prevertices is the accessory-parameter problem and is generally nonlinear. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.25. State and apply the central principle behind Normalization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Three real degrees of freedom can be fixed by half-plane Möbius automorphisms. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.26. State and apply the central principle behind Rectangles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Symmetry reduces rectangle maps to elliptic integrals. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 18.27. State and apply the central principle behind Slits and degenerate polygons. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Limiting polygon configurations produce maps to slit domains and strips. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Part IV

Special Functions

Chapter 19

The Gamma Function

The Gamma function extends factorials to the complex plane through an integral, functional equation, and analytic continuation. It is the prototype of a special function whose poles and growth encode arithmetic structure.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

19.1 Euler integral

For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$.

19.2 Convergence

Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power.

19.3 Functional equation

Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$.

19.4 Factorials

For positive integers, $\Gamma(n+1) = n!$.

19.5 Analyticity in the half-plane

Differentiation under the integral sign gives holomorphic dependence on s for $\Re s > 0$.

19.6 Meromorphic continuation

The functional equation extends Γ to the plane with simple poles at nonpositive integers.

19.7 Residues at poles

Pole residues are determined recursively from the residue at zero.

19.8 Logarithmic derivative

The digamma function $\psi = \Gamma'/\Gamma$ converts products and functional identities into additive formulas.

19.9 Growth preview

Stirling asymptotics describe factorial-like growth in sectors.

19.10 Core structural results

Theorem 19.1 (Gamma functional equation)**.** *For* $\Re s > 0$*,* $\Gamma(s+1) = s\Gamma(s)$*.*

Proof. Integrate $\int_0^\infty t^s e^{-t} dt$ by parts. Boundary terms vanish because $t^s e^{-t}$ decays at infinity and tends to zero at the origin under the real-part condition. $\square$

Theorem 19.2 (Meromorphic continuation)**.** *The functional equation extends* Γ *meromorphically to* $\mathbb{C}$ *with simple poles at* $0, -1, -2, \ldots$*.*

Proof. For any N, define $\Gamma(s) = \Gamma(s+N)/(s(s+1)\cdots(s+N-1))$ where the numerator integral is holomorphic for $\Re(s+N) > 0$. These local formulas agree on overlaps by the functional equation. $\square$

Theorem 19.3 (Residue formula)**.** *At* $s = -n$*,* $n \geq 0$*,* $\mathrm{Res}(\Gamma, -n) = (-1)^n/n!$*.*

Proof. Start from $\mathrm{Res}(\Gamma, 0) = 1$ using $\Gamma(s+1) = s\Gamma(s)$ and propagate residues recursively through the functional equation. $\square$

19.11 Worked examples

Example 19.4 (Euler integral diagnostic)**.** Give a rigorous worked analysis of Euler integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 19.5 (Convergence diagnostic)**.** Give a rigorous worked analysis of Convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 19.6 (Functional equation diagnostic)**.** Give a rigorous worked analysis of Functional equation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 19.7 (Factorials diagnostic)**.** Give a rigorous worked analysis of Factorials. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For positive integers, $\Gamma(n+1) = n!$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

19.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 19.8 (Euler integral diagnostic)**.** Give a rigorous worked analysis of Euler integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 19.9 (Convergence diagnostic)**.** Give a rigorous worked analysis of Convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 19.10 (Functional equation diagnostic)**.** Give a rigorous worked analysis of Functional equation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 19.11 (Factorials diagnostic)**.** Give a rigorous worked analysis of Factorials. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For positive integers, $\Gamma(n+1) = n!$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 19.12 (Analyticity in the half-plane diagnostic)**.** Give a rigorous worked analysis of Analyticity in the half-plane. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiation under the integral sign gives holomorphic dependence on s for $\Re s > 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 19.13 (Meromorphic continuation diagnostic)**.** Give a rigorous worked analysis of Meromorphic continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The functional equation extends Γ to the plane with simple poles at nonpositive integers. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.14 (Residues at poles diagnostic)**.** Give a rigorous worked analysis of Residues at poles. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Pole residues are determined recursively from the residue at zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.15 (Logarithmic derivative diagnostic)**.** Give a rigorous worked analysis of Logarithmic derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The digamma function $\psi = \Gamma'/\Gamma$ converts products and functional identities into additive formulas. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 19.16 (Gamma functional equation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $\Re s > 0$, $\Gamma(s+1) = s\Gamma(s)$.

Solution. Integrate $\int_0^\infty t^s e^{-t} dt$ by parts. Boundary terms vanish because $t^s e^{-t}$ decays at infinity and tends to zero at the origin under the real-part condition. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 19.17 (Meromorphic continuation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The functional equation extends Γ meromorphically to $\mathbb{C}$ with simple poles at $0, -1, -2, \ldots$.

Solution. For any N, define $\Gamma(s) = \Gamma(s+N)/(s(s+1)\cdots(s+N-1))$ where the numerator integral is holomorphic for $\Re(s+N) > 0$. These local formulas agree on overlaps by the functional equation. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 19.18 (Residue formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: At $s = -n$, $n \geq 0$, $\mathrm{Res}(\Gamma, -n) = (-1)^n/n!$.

Solution. Start from $\mathrm{Res}(\Gamma, 0) = 1$ using $\Gamma(s+1) = s\Gamma(s)$ and propagate residues recursively through the functional equation. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 19.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Gamma function extends factorials to the complex plane through an integral, functional equation, and analytic continuation. It is the prototype of a special function whose poles and growth encode arithmetic structure. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

19.13 Exercises with complete solutions

Exercise 19.20. State and apply the central principle behind Euler integral. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. For $\Re s > 0$, define $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}dt$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.21. State and apply the central principle behind Convergence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Near zero the condition $\Re s > 0$ controls t^{s-1}; at infinity exponential decay dominates every power. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.22. State and apply the central principle behind Functional equation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.23. State and apply the central principle behind Factorials. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. For positive integers, $\Gamma(n+1) = n!$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.24. State and apply the central principle behind Analyticity in the half-plane. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Differentiation under the integral sign gives holomorphic dependence on s for $\Re s > 0$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.25. State and apply the central principle behind Meromorphic continuation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The functional equation extends Γ to the plane with simple poles at nonpositive integers. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 19.26. State and apply the central principle behind Residues at poles. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Pole residues are determined recursively from the residue at zero. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 19.27. State and apply the central principle behind Logarithmic derivative. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The digamma function $\psi = \Gamma'/\Gamma$ converts products and functional identities into additive formulas. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 20

Beta and Gamma Identities

The Beta function packages a two-endpoint integral and connects directly to Gamma through a change of variables. Their identities provide a compact calculus for factorials, binomial coefficients, trigonometric integrals, and parameter differentiation.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

20.1 Beta integral

For $\Re x, \Re y > 0$, $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$.

20.2 Symmetry

The substitution $t \mapsto 1-t$ gives $B(x,y) = B(y,x)$.

20.3 Gamma relation

A two-dimensional change of variables yields $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$.

20.4 Recurrences

Integration by parts and the Gamma relation produce rational recurrences in the parameters.

20.5 Trigonometric form

The substitution $t = \sin^2\theta$ gives Beta evaluations of sine and cosine powers.

20.6 Factorial identities

At integer and half-integer arguments, Beta and Gamma produce binomial and central-binomial formulas.

20.7 Parameter derivatives

Differentiation with respect to parameters introduces logarithmic factors inside the integral.

20.8 Duplication preview

Combining Beta identities yields the Gamma duplication formula.

20.9 Analytic continuation

The Gamma quotient extends Beta meromorphically beyond its defining integral region.

20.10 Core structural results

Theorem 20.1 (Beta–Gamma identity)**.** *For* $\Re x, \Re y > 0$*,* $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x + y)$*.*

Proof. Write the product of Gamma integrals over the first quadrant and substitute $u = rt$, $v = r(1 - t)$ with $r > 0$, $0 < t < 1$. The Jacobian is r, and the integral factors into $\Gamma(x + y)B(x, y)$. ☐

Theorem 20.2 (Trigonometric Beta formula)**.** *For* $a, b > 0$*,* $\int_0^{\pi/2} \sin^{a-1}\theta \cos^{b-1}\theta \, d\theta = \frac{1}{2}B(a/2, b/2)$*.*

Proof. Set $t = \sin^2\theta$, so $dt = 2\sin\theta\cos\theta \, d\theta$ and collect the resulting powers. ☐

Theorem 20.3 (Beta recurrence)**.** $B(x + 1, y) = \frac{x}{x+y}B(x, y)$ *in the common domain of definition.*

Proof. Use the Beta–Gamma identity and the Gamma functional equation in numerator and denominator. ☐

20.11 Worked examples

Example 20.4 (Beta integral diagnostic)**.** Give a rigorous worked analysis of Beta integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For $\Re x, \Re y > 0$, $B(x, y) = \int_0^1 t^{x-1}(1 - t)^{y-1}dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 20.5 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The substitution $t \mapsto 1 - t$ gives $B(x, y) = B(y, x)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 20.6 (Gamma relation diagnostic)**.** Give a rigorous worked analysis of Gamma relation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A two-dimensional change of variables yields $B(x, y) =$

$\Gamma(x)\Gamma(y)/\Gamma(x+y)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 20.7 (Recurrences diagnostic)**.** Give a rigorous worked analysis of Recurrences. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Integration by parts and the Gamma relation produce rational recurrences in the parameters. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

20.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 20.8 (Beta integral diagnostic)**.** Give a rigorous worked analysis of Beta integral. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For $\Re x, \Re y > 0$, $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.9 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The substitution $t \mapsto 1-t$ gives $B(x,y) = B(y,x)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.10 (Gamma relation diagnostic)**.** Give a rigorous worked analysis of Gamma relation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A two-dimensional change of variables yields $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.11 (Recurrences diagnostic)**.** Give a rigorous worked analysis of Recurrences. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Integration by parts and the Gamma relation produce rational recurrences in the parameters. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.12 (Trigonometric form diagnostic)**.** Give a rigorous worked analysis of Trigonometric form. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The substitution $t = \sin^2\theta$ gives Beta evaluations of sine and cosine powers. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 20.13 (Factorial identities diagnostic)**.** Give a rigorous worked analysis of Factorial identities. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. At integer and half-integer arguments, Beta and Gamma produce binomial and central-binomial formulas. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 20.14 (Parameter derivatives diagnostic)**.** Give a rigorous worked analysis of Parameter derivatives. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiation with respect to parameters introduces logarithmic factors inside the integral. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 20.15 (Duplication preview diagnostic)**.** Give a rigorous worked analysis of Duplication preview. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Combining Beta identities yields the Gamma duplication formula. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 20.16 (Beta–Gamma identity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $\Re x, \Re y > 0$, $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$.

Solution. Write the product of Gamma integrals over the first quadrant and substitute $u = rt$, $v = r(1-t)$ with $r > 0$, $0 < t < 1$. The Jacobian is r, and the integral factors into $\Gamma(x+y)B(x, y)$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 20.17 (Trigonometric Beta formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $a, b > 0$, $\int_0^{\pi/2} \sin^{a-1}\theta \cos^{b-1}\theta \, d\theta = \frac{1}{2}B(a/2, b/2)$.

Solution. Set $t = \sin^2\theta$, so $dt = 2\sin\theta\cos\theta \, d\theta$ and collect the resulting powers. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 20.18 (Beta recurrence proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: $B(x+1, y) = \frac{x}{x+y}B(x, y)$ in the common domain of definition.

Solution. Use the Beta–Gamma identity and the Gamma functional equation in numerator and denominator. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 20.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Beta function packages a two-endpoint integral and connects directly to Gamma through a change of variables. Their identities provide a compact calculus for factorials, binomial coefficients, trigonometric integrals, and parameter differentiation. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

20.13 Exercises with complete solutions

Exercise 20.20. State and apply the central principle behind Beta integral. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For $\Re x, \Re y > 0$, $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.21. State and apply the central principle behind Symmetry. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The substitution $t \mapsto 1-t$ gives $B(x,y) = B(y,x)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.22. State and apply the central principle behind Gamma relation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A two-dimensional change of variables yields $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.23. State and apply the central principle behind Recurrences. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Integration by parts and the Gamma relation produce rational recurrences in the parameters. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.24. State and apply the central principle behind Trigonometric form. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The substitution $t = \sin^2\theta$ gives Beta evaluations of sine and cosine powers. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.25. State and apply the central principle behind Factorial identities. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. At integer and half-integer arguments, Beta and Gamma produce binomial and central-binomial formulas. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.26. State and apply the central principle behind Parameter derivatives. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Differentiation with respect to parameters introduces logarithmic factors inside the integral. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 20.27. State and apply the central principle behind Duplication preview. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Combining Beta identities yields the Gamma duplication formula. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 21

Keyhole Contours and Branch-Cut Integrals

Keyhole contours turn branch discontinuities into computable factors. They are the standard mechanism for integrals involving fractional powers, logarithms, and branch cuts.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

21.1 Branch choice

A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen.

21.2 Keyhole geometry

A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles.

21.3 Jump factor

The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut.

21.4 Vanishing circular arcs

Power estimates determine whether the small and large circular contributions disappear.

21.5 Residue contribution

Poles away from the branch cut contribute through the residue theorem.

21.6 Logarithmic integrals

Differentiating branch-power identities with respect to exponents produces logarithms.

21.7 Orientation bookkeeping

Upper and lower sides of the cut have opposite traversal directions.

21.8 Parameter windows

The allowed real parts of exponents are dictated by convergence at zero and infinity.

21.9 Contour variants

Hankel contours and wedge contours implement the same jump principle for other branch geometries.

21.10 Core structural results

Theorem 21.1 (Keyhole jump identity)**.** *For a branch cut along the positive real axis with argument in* $(0, 2\pi)$*, the upper and lower boundary values of* z^α *differ by the factor* $e^{2\pi i\alpha}$*.*

Proof. On the upper bank the argument tends to 0, while on the lower bank it tends to 2π. Substitution into $e^{\alpha(\log r + i \arg z)}$ gives the factor. □

Theorem 21.2 (Keyhole reduction principle)**.** *If the circular arcs vanish, the keyhole contour integral reduces to a scalar jump factor times the corresponding real integral along the cut.*

Proof. Write the contour as upper bank plus outer circle plus lower bank plus inner circle. Let the radii tend to their limits; the circle contributions vanish and the two bank integrals combine using their branch-value relation and opposite orientations. □

Theorem 21.3 (Differentiation in exponent)**.** *When a keyhole identity is locally uniformly convergent in a parameter* α*, differentiating it with respect to* α *introduces factors of* $\operatorname{Log} z$ *and yields logarithmic integral identities.*

Proof. Differentiate $z^\alpha = e^{\alpha \operatorname{Log} z}$ under the integral sign, justified by a common integrable majorant on a compact parameter subregion. □

21.11 Worked examples

Example 21.4 (Branch choice diagnostic)**.** Give a rigorous worked analysis of Branch choice. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 21.5 (Keyhole geometry diagnostic)**.** Give a rigorous worked analysis of Keyhole geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 21.6 (Jump factor diagnostic)**.** Give a rigorous worked analysis of Jump factor. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 21.7 (Vanishing circular arcs diagnostic)**.** Give a rigorous worked analysis of Vanishing circular arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Power estimates determine whether the small and large circular contributions disappear. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

21.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 21.8 (Branch choice diagnostic)**.** Give a rigorous worked analysis of Branch choice. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.9 (Keyhole geometry diagnostic)**.** Give a rigorous worked analysis of Keyhole geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.10 (Jump factor diagnostic)**.** Give a rigorous worked analysis of Jump factor. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.11 (Vanishing circular arcs diagnostic)**.** Give a rigorous worked analysis of Vanishing circular arcs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Power estimates determine whether the small and large circular contributions disappear. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.12 (Residue contribution diagnostic)**.** Give a rigorous worked analysis of Residue contribution. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Poles away from the branch cut contribute through the residue theorem. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.13 (Logarithmic integrals diagnostic)**.** Give a rigorous worked analysis of Logarithmic integrals. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differentiating branch-power identities with respect to exponents produces logarithms. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.14 (Orientation bookkeeping diagnostic)**.** Give a rigorous worked analysis of Orientation bookkeeping. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Upper and lower sides of the cut have opposite traversal directions. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.15 (Parameter windows diagnostic)**.** Give a rigorous worked analysis of Parameter windows. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The allowed real parts of exponents are dictated by convergence at zero and infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 21.16 (Keyhole jump identity proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a branch cut along the positive real axis with argument in $(0, 2\pi)$, the upper and lower boundary values of z^α differ by the factor $e^{2\pi i\alpha}$.

Solution. On the upper bank the argument tends to 0, while on the lower bank it tends to 2π. Substitution into $e^{\alpha(\log r + i \arg z)}$ gives the factor. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 21.17 (Keyhole reduction principle proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If the circular arcs vanish, the keyhole contour integral reduces to a scalar jump factor times the corresponding real integral along the cut.

Solution. Write the contour as upper bank plus outer circle plus lower bank plus inner circle. Let the radii tend to their limits; the circle contributions vanish and the two bank integrals combine using their branch-value relation and opposite orientations. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 21.18 (Differentiation in exponent proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: When a keyhole identity is locally uniformly convergent in a parameter α, differentiating it with respect to α introduces factors of $\operatorname{Log} z$ and yields logarithmic integral identities.

Solution. Differentiate $z^\alpha = e^{\alpha \operatorname{Log} z}$ under the integral sign, justified by a common integrable majorant on a compact parameter subregion. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 21.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Keyhole contours turn branch discontinuities into computable factors. They are the standard mechanism for integrals involving fractional powers, logarithms, and branch cuts. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

21.13 Exercises with complete solutions

Exercise 21.20. State and apply the central principle behind Branch choice. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A branch of $z^\alpha = e^{\alpha \operatorname{Log} z}$ must be fixed before the contour is chosen. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 21.21. State and apply the central principle behind Keyhole geometry. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A keyhole contour runs along the two sides of a branch cut and connects them by small and large circles. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 21.22. State and apply the central principle behind Jump factor. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The two boundary values of z^α differ by a phase $e^{2\pi i\alpha}$ across the cut. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 21.23. State and apply the central principle behind Vanishing circular arcs. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Power estimates determine whether the small and large circular contributions disappear. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 21.24. State and apply the central principle behind Residue contribution. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Poles away from the branch cut contribute through the residue theorem. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 21.25. State and apply the central principle behind Logarithmic integrals. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Differentiating branch-power identities with respect to exponents produces logarithms. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 21.26. State and apply the central principle behind Orientation bookkeeping. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Upper and lower sides of the cut have opposite traversal directions. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 21.27. State and apply the central principle behind Parameter windows. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The allowed real parts of exponents are dictated by convergence at zero and infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Part V

Riemann Surfaces

Chapter 22

From Analytic Continuation to Riemann Surfaces

Analytic continuation naturally leads beyond planar single-valued functions. A Riemann surface is the geometric object on which a locally holomorphic multivalued expression becomes an ordinary single-valued holomorphic function.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

22.1 Local branches

Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches.

22.2 Branch incompatibility

Different continuations around loops can disagree after returning to the same plane point.

22.3 Surface of germs

Collecting compatible germs over plane points separates distinct analytic branches.

22.4 Projection map

A Riemann surface often comes with a holomorphic projection to the plane or sphere.

22.5 Charts

Local coordinates make the surface look like open subsets of $\mathbb{C}$.

22.6 Holomorphic functions on surfaces

Holomorphicity is defined through compatible coordinate charts.

22.7 Branch points

Projection can fail to be locally one-to-one at ramification points.

22.8 Maximal analytic continuation

The maximal continuation of a germ can be organized as a connected Riemann surface.

22.9 Examples

The logarithm surface and the square-root surface show infinite-sheeted and two-sheeted behavior.

22.10 Core structural results

Theorem 22.1 (Local single-valuedness)**.** *Every point of a Riemann surface has a neighborhood on which a holomorphic projection that is locally biholomorphic can be used as a complex coordinate.*

Proof. This is built into the chart structure: a local coordinate is a homeomorphism to a plane domain with holomorphic transition maps. □

Theorem 22.2 (Germ-surface principle)**.** *Compatible analytic continuations of a germ can be organized into a set of germs with a natural projection to the base point and a local complex coordinate given by that projection.*

Proof. Around each germ choose a representative on a small disk. Nearby points correspond to the germs of the same representative; these neighborhoods provide charts and compatible transition maps. □

Theorem 22.3 (Single-valued lift)**.** *A multivalued analytic expression becomes single-valued when evaluated on its surface of analytic germs.*

Proof. A point of the surface records not just the base coordinate but the chosen local germ, so evaluation of that germ is unambiguous. □

22.11 Worked examples

Example 22.4 (Local branches diagnostic)**.** Give a rigorous worked analysis of Local branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 22.5 (Branch incompatibility diagnostic)**.** Give a rigorous worked analysis of Branch incompatibility. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Different continuations around loops can disagree after returning to the same plane point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 22.6 (Surface of germs diagnostic)**.** Give a rigorous worked analysis of Surface of germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Collecting compatible germs over plane points separates distinct analytic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 22.7 (Projection map diagnostic)**.** Give a rigorous worked analysis of Projection map. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A Riemann surface often comes with a holomorphic projection to the plane or sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

22.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 22.8 (Local branches diagnostic)**.** Give a rigorous worked analysis of Local branches. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.9 (Branch incompatibility diagnostic)**.** Give a rigorous worked analysis of Branch incompatibility. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Different continuations around loops can disagree after returning to the same plane point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.10 (Surface of germs diagnostic)**.** Give a rigorous worked analysis of Surface of germs. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Collecting compatible germs over plane points separates distinct analytic branches. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.11 (Projection map diagnostic)**.** Give a rigorous worked analysis of Projection map. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A Riemann surface often comes with a holomorphic projection to the plane or sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.12 (Charts diagnostic)**.** Give a rigorous worked analysis of Charts. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Local coordinates make the surface look like open subsets of $\mathbb{C}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.13 (Holomorphic functions on surfaces diagnostic)**.** Give a rigorous worked analysis of Holomorphic functions on surfaces. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Holomorphicity is defined through compatible coordinate charts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.14 (Branch points diagnostic)**.** Give a rigorous worked analysis of Branch points. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Projection can fail to be locally one-to-one at ramification points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.15 (Maximal analytic continuation diagnostic)**.** Give a rigorous worked analysis of Maximal analytic continuation. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The maximal continuation of a germ can be organized as a connected Riemann surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 22.16 (Local single-valuedness proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every point of a Riemann surface has a neighborhood on which a holomorphic projection that is locally biholomorphic can be used as a complex coordinate.

Solution. This is built into the chart structure: a local coordinate is a homeomorphism to a plane domain with holomorphic transition maps. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 22.17 (Germ-surface principle proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Compatible analytic continuations of a germ can be organized into a set of germs with a natural projection to the base point and a local complex coordinate given by that projection.

Solution. Around each germ choose a representative on a small disk. Nearby points correspond to the germs of the same representative; these neighborhoods provide charts and compatible transition maps. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 22.18 (Single-valued lift proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A multivalued analytic expression becomes single-valued when evaluated on its surface of analytic germs.

Solution. A point of the surface records not just the base coordinate but the chosen local germ, so evaluation of that germ is unambiguous. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 22.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Analytic continuation naturally leads beyond planar single-valued functions. A Riemann surface is the geometric object on which a locally holomorphic multivalued expression becomes an ordinary single-valued holomorphic function. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

22.13 Exercises with complete solutions

Exercise 22.20. State and apply the central principle behind Local branches. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Square roots, logarithms, and inverse functions have locally single-valued holomorphic branches. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 22.21. State and apply the central principle behind Branch incompatibility. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Different continuations around loops can disagree after returning to the same plane point. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 22.22. State and apply the central principle behind Surface of germs. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Collecting compatible germs over plane points separates distinct analytic branches. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 22.23. State and apply the central principle behind Projection map. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A Riemann surface often comes with a holomorphic projection to the plane or sphere. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 22.24. State and apply the central principle behind Charts. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Local coordinates make the surface look like open subsets of $\mathbb{C}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.25. State and apply the central principle behind Holomorphic functions on surfaces. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Holomorphicity is defined through compatible coordinate charts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.26. State and apply the central principle behind Branch points. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Projection can fail to be locally one-to-one at ramification points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 22.27. State and apply the central principle behind Maximal analytic continuation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The maximal continuation of a germ can be organized as a connected Riemann surface. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 23

Covering Maps and Monodromy

Covering maps formalize the idea of many sheets lying over one base domain. Path lifting and monodromy translate topology of loops into permutations of analytic branches.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

23.1 Covering definition

A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods.

23.2 Sheets

Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood.

23.3 Path lifting

A path in the base has a unique lift once its starting point in the fiber is chosen.

23.4 Loop lifting

A loop may lift to a path whose endpoint is a different point of the same fiber.

23.5 Monodromy action

Homotopy classes of loops act by permutations on the fiber.

23.6 Deck transformations

Automorphisms of the covering commute with the projection and permute sheets globally.

23.7 Exponential covering

$z \mapsto e^z$ covers $\mathbb{C}^*$ with deck translations by $2\pi i\mathbb{Z}$.

23.8 Power maps

$z \mapsto z^n$ covers $\mathbb{C}^*$ by itself with cyclic monodromy.

23.9 Analytic continuation

Branches of inverse functions move along lifted paths, making covering theory a natural language for continuation without branch points.

23.10 Core structural results

Theorem 23.1 (Unique path lifting). *For a covering $p : X \to Y$, any path γ in Y and any $x_0 \in p^{-1}(\gamma(0))$ have a unique lift starting at x_0.*

Proof. Cover the compact parameter interval by subintervals mapping into evenly covered neighborhoods and construct the lift step by step. Local uniqueness on each sheet forces global uniqueness. ☐

Theorem 23.2 (Homotopy invariance of monodromy). *Homotopic loops with fixed base point induce the same permutation of the fiber.*

Proof. Use the homotopy lifting property for coverings. The lifted endpoints vary continuously in a discrete fiber, hence remain constant through the homotopy. ☐

Theorem 23.3 (Exponential covering). *The exponential map $\exp : \mathbb{C} \to \mathbb{C}^*$ is a covering map with deck group $z \mapsto z + 2\pi ik$.*

Proof. Small disks in $\mathbb{C}^*$ avoiding zero admit logarithm branches; their inverse images are disjoint translates of one logarithm branch, and exponential is biholomorphic on each translate. ☐

23.11 Worked examples

Example 23.4 (Covering definition diagnostic). Give a rigorous worked analysis of Covering definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 23.5 (Sheets diagnostic). Give a rigorous worked analysis of Sheets. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 23.6 (Path lifting diagnostic)**.** Give a rigorous worked analysis of Path lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A path in the base has a unique lift once its starting point in the fiber is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 23.7 (Loop lifting diagnostic)**.** Give a rigorous worked analysis of Loop lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A loop may lift to a path whose endpoint is a different point of the same fiber. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

23.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 23.8 (Covering definition diagnostic)**.** Give a rigorous worked analysis of Covering definition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.9 (Sheets diagnostic)**.** Give a rigorous worked analysis of Sheets. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.10 (Path lifting diagnostic)**.** Give a rigorous worked analysis of Path lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A path in the base has a unique lift once its starting point in the fiber is chosen. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.11 (Loop lifting diagnostic)**.** Give a rigorous worked analysis of Loop lifting. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A loop may lift to a path whose endpoint is a different point of the same fiber. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.12 (Monodromy action diagnostic)**.** Give a rigorous worked analysis of Monodromy action. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Homotopy classes of loops act by permutations on the fiber. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.13 (Deck transformations diagnostic)**.** Give a rigorous worked analysis of Deck transformations. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Automorphisms of the covering commute with the projection and permute sheets globally. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.14 (Exponential covering diagnostic)**.** Give a rigorous worked analysis of Exponential covering. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $z \mapsto e^z$ covers $\mathbb{C}^*$ with deck translations by $2\pi i\mathbb{Z}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.15 (Power maps diagnostic)**.** Give a rigorous worked analysis of Power maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. $z \mapsto z^n$ covers $\mathbb{C}^*$ by itself with cyclic monodromy. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 23.16 (Unique path lifting proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a covering $p : X \to Y$, any path γ in Y and any $x_0 \in p^{-1}(\gamma(0))$ have a unique lift starting at x_0.

Solution. Cover the compact parameter interval by subintervals mapping into evenly covered neighborhoods and construct the lift step by step. Local uniqueness on each sheet forces global uniqueness. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 23.17 (Homotopy invariance of monodromy proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Homotopic loops with fixed base point induce the same permutation of the fiber.

Solution. Use the homotopy lifting property for coverings. The lifted endpoints vary continuously in a discrete fiber, hence remain constant through the homotopy. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 23.18 (Exponential covering proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The exponential map $\exp : \mathbb{C} \to \mathbb{C}^*$ is a covering map with deck group $z \mapsto z + 2\pi i k$.

Solution. Small disks in $\mathbb{C}^*$ avoiding zero admit logarithm branches; their inverse images are disjoint translates of one logarithm branch, and exponential is biholomorphic on each translate. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 23.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Covering maps formalize the idea of many sheets lying over one base domain. Path lifting and monodromy translate topology of loops into permutations of analytic branches. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

23.13 Exercises with complete solutions

Exercise 23.20. State and apply the central principle behind Covering definition. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A covering map is locally a disjoint union of homeomorphic sheets over evenly covered neighborhoods. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 23.21. State and apply the central principle behind Sheets. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Each component over an evenly covered neighborhood projects homeomorphically to the neighborhood. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 23.22. State and apply the central principle behind Path lifting. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A path in the base has a unique lift once its starting point in the fiber is chosen. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 23.23. State and apply the central principle behind Loop lifting. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A loop may lift to a path whose endpoint is a different point of the same fiber. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 23.24. State and apply the central principle behind Monodromy action. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Homotopy classes of loops act by permutations on the fiber. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 23.25. State and apply the central principle behind Deck transformations. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Automorphisms of the covering commute with the projection and permute sheets globally. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.26. State and apply the central principle behind Exponential covering. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. $z \mapsto e^z$ covers $\mathbb{C}^*$ with deck translations by $2\pi i\mathbb{Z}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 23.27. State and apply the central principle behind Power maps. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. $z \mapsto z^n$ covers $\mathbb{C}^*$ by itself with cyclic monodromy. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 24

Branched Coverings

Branched coverings relax ordinary covering behavior at isolated ramification points. Locally they are modeled by power maps, and the ramification indices encode how sheets merge.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

24.1 Branched covering

Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge.

24.2 Local power model

Near a ramification point, suitable coordinates often put the map in form $w = z^e$.

24.3 Ramification index

The integer e measures local multiplicity; $e > 1$ means ramification.

24.4 Branch values

Images of ramification points are branch values in the base.

24.5 Square-root surface

The surface $w^2 = z$ is a two-sheeted cover of the sphere branched at zero and infinity.

24.6 Polynomial maps

A polynomial viewed on the sphere gives a branched covering whose critical points are ramification points.

24.7 Monodromy around branch values

Loops around branch values permute sheets nontrivially.

24.8 Degree

Away from branch values, every fiber has the same number of points counted with multiplicity.

24.9 Riemann–Hurwitz preview

Total ramification contributes to the genus relation between compact surfaces.

24.10 Core structural results

Theorem 24.1 (Local ramification model)**.** *A nonconstant holomorphic map between Riemann surfaces has local coordinates in which it is $z \mapsto z^e$ for a unique integer $e \geq 1$.*

Proof. Expand the map in a local coordinate around the source point, factor out the first nonzero term of order e, and absorb the nonvanishing holomorphic factor into a new local coordinate using a holomorphic eth root. ☐

Theorem 24.2 (Critical-point criterion)**.** *A point is ramified iff the derivative of a local coordinate representation vanishes there.*

Proof. In the local model z^e, the derivative at zero is nonzero exactly when $e = 1$. ☐

Theorem 24.3 (Constant generic degree)**.** *For a proper finite branched covering, the sum of ramification multiplicities in a fiber is constant away from degenerations and equals the degree.*

Proof. Locally, zeros of the equation $p(x) = y$ are counted with multiplicity; the argument principle makes this count stable as y moves without crossing branch values. ☐

24.11 Worked examples

Example 24.4 (Branched covering diagnostic)**.** Give a rigorous worked analysis of Branched covering. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 24.5 (Local power model diagnostic)**.** Give a rigorous worked analysis of Local power model. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Near a ramification point, suitable coordinates often put the map in form $w = z^e$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 24.6 (Ramification index diagnostic). Give a rigorous worked analysis of Ramification index. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The integer e measures local multiplicity; $e > 1$ means ramification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 24.7 (Branch values diagnostic). Give a rigorous worked analysis of Branch values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Images of ramification points are branch values in the base. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

24.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 24.8 (Branched covering diagnostic). Give a rigorous worked analysis of Branched covering. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 24.9 (Local power model diagnostic). Give a rigorous worked analysis of Local power model. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Near a ramification point, suitable coordinates often put the map in form $w = z^e$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 24.10 (Ramification index diagnostic). Give a rigorous worked analysis of Ramification index. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The integer e measures local multiplicity; $e > 1$ means ramification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 24.11 (Branch values diagnostic). Give a rigorous worked analysis of Branch values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Images of ramification points are branch values in the base. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 24.12 (Square-root surface diagnostic). Give a rigorous worked analysis of Square-root surface. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The surface $w^2 = z$ is a two-sheeted cover of the sphere branched at zero and infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.13 (Polynomial maps diagnostic)**.** Give a rigorous worked analysis of Polynomial maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A polynomial viewed on the sphere gives a branched covering whose critical points are ramification points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.14 (Monodromy around branch values diagnostic)**.** Give a rigorous worked analysis of Monodromy around branch values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Loops around branch values permute sheets nontrivially. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.15 (Degree diagnostic)**.** Give a rigorous worked analysis of Degree. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Away from branch values, every fiber has the same number of points counted with multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 24.16 (Local ramification model proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A nonconstant holomorphic map between Riemann surfaces has local coordinates in which it is $z \mapsto z^e$ for a unique integer $e \geq 1$.

Solution. Expand the map in a local coordinate around the source point, factor out the first nonzero term of order e, and absorb the nonvanishing holomorphic factor into a new local coordinate using a holomorphic eth root. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 24.17 (Critical-point criterion proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A point is ramified iff the derivative of a local coordinate representation vanishes there.

Solution. In the local model z^e, the derivative at zero is nonzero exactly when $e = 1$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 24.18 (Constant generic degree proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a proper finite branched covering, the sum of ramification multiplicities in a fiber is constant away from degenerations and equals the degree.

Solution. Locally, zeros of the equation $p(x) = y$ are counted with multiplicity; the argument principle makes this count stable as y moves without crossing branch values. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 24.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Branched coverings relax ordinary covering behavior at isolated ramification points. Locally they are modeled by power maps, and the ramification indices encode how sheets merge. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

24.13 Exercises with complete solutions

Exercise 24.20. State and apply the central principle behind Branched covering. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Away from a discrete branch set, the map is an ordinary covering; over branch points sheets merge. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 24.21. State and apply the central principle behind Local power model. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Near a ramification point, suitable coordinates often put the map in form $w = z^e$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 24.22. State and apply the central principle behind Ramification index. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The integer e measures local multiplicity; $e > 1$ means ramification. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 24.23. State and apply the central principle behind Branch values. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Images of ramification points are branch values in the base. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 24.24. State and apply the central principle behind Square-root surface. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The surface $w^2 = z$ is a two-sheeted cover of the sphere branched at zero and infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 24.25. State and apply the central principle behind Polynomial maps. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A polynomial viewed on the sphere gives a branched covering whose critical points are ramification points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.26. State and apply the central principle behind Monodromy around branch values. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Loops around branch values permute sheets nontrivially. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 24.27. State and apply the central principle behind Degree. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Away from branch values, every fiber has the same number of points counted with multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 25

Construction by Gluing

Gluing constructs Riemann surfaces by identifying open pieces through holomorphic transition maps. This is the global counterpart of defining a manifold from compatible charts.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

25.1 Gluing data

Start with complex domains and specify identifications between open subsets.

25.2 Transition maps

Identifications must be biholomorphic and satisfy inverse and cocycle compatibility.

25.3 Quotient space

The disjoint union modulo the identifications becomes the candidate surface.

25.4 Hausdorff condition

Gluing must be controlled so the quotient remains Hausdorff.

25.5 Charts from pieces

Each original domain descends locally to a coordinate chart on the quotient.

25.6 Sphere from two charts

Two copies of the plane glued by $w = 1/z$ on punctured regions produce the Riemann sphere.

25.7 Algebraic curves

Local branches of an algebraic relation can be glued into a smooth Riemann surface.

25.8 Cut-and-paste constructions

Copies of slit planes or polygons can be glued to produce root surfaces and tori.

25.9 Transition-cocycle viewpoint

The global complex structure is completely encoded by holomorphic chart transition functions.

25.10 Core structural results

Theorem 25.1 (Holomorphic gluing theorem)**.** *If Hausdorff quotient gluing data have biholomorphic transition maps satisfying the cocycle conditions, the quotient inherits a unique Riemann-surface structure for which the piece maps are local charts.*

Proof. Use the images of the original domains as charts. On overlaps their transition maps are exactly the prescribed biholomorphisms, so the atlas is compatible and defines the complex structure. □

Theorem 25.2 (Sphere gluing)**.** *Gluing two copies of* $\mathbb{C}$ *by* $w = 1/z$ *on* $\mathbb{C}^*$ *produces a compact Riemann surface biholomorphic to* $\widehat{\mathbb{C}}$*.*

Proof. Map one chart to finite complex numbers and the origin of the second chart to infinity. The transition relation is precisely the standard coordinate change at infinity. □

Theorem 25.3 (Cocycle consistency)**.** *For triple overlaps, transition maps must satisfy* $g_{ij} \circ g_{jk} = g_{ik}$*.*

Proof. A point represented through three charts must have the same coordinate identification independent of the path through the overlap graph; the cocycle condition is exactly that independence. □

25.11 Worked examples

Example 25.4 (Gluing data diagnostic)**.** Give a rigorous worked analysis of Gluing data. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Start with complex domains and specify identifications between open subsets. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 25.5 (Transition maps diagnostic)**.** Give a rigorous worked analysis of Transition maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Identifications must be biholomorphic and satisfy inverse and cocycle compatibility. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 25.6 (Quotient space diagnostic)**.** Give a rigorous worked analysis of Quotient space. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The disjoint union modulo the identifications becomes the candidate surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 25.7 (Hausdorff condition diagnostic)**.** Give a rigorous worked analysis of Hausdorff condition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Gluing must be controlled so the quotient remains Hausdorff. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

25.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 25.8 (Gluing data diagnostic)**.** Give a rigorous worked analysis of Gluing data. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Start with complex domains and specify identifications between open subsets. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.9 (Transition maps diagnostic)**.** Give a rigorous worked analysis of Transition maps. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Identifications must be biholomorphic and satisfy inverse and cocycle compatibility. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.10 (Quotient space diagnostic)**.** Give a rigorous worked analysis of Quotient space. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The disjoint union modulo the identifications becomes the candidate surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.11 (Hausdorff condition diagnostic)**.** Give a rigorous worked analysis of Hausdorff condition. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Gluing must be controlled so the quotient remains Hausdorff. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 25.12 (Charts from pieces diagnostic)**.** Give a rigorous worked analysis of Charts from pieces. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Each original domain descends locally to a coordinate chart on the quotient. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 25.13 (Sphere from two charts diagnostic)**.** Give a rigorous worked analysis of Sphere from two charts. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Two copies of the plane glued by $w = 1/z$ on punctured regions produce the Riemann sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 25.14 (Algebraic curves diagnostic)**.** Give a rigorous worked analysis of Algebraic curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Local branches of an algebraic relation can be glued into a smooth Riemann surface. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 25.15 (Cut-and-paste constructions diagnostic)**.** Give a rigorous worked analysis of Cut-and-paste constructions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Copies of slit planes or polygons can be glued to produce root surfaces and tori. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 25.16 (Holomorphic gluing theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If Hausdorff quotient gluing data have biholomorphic transition maps satisfying the cocycle conditions, the quotient inherits a unique Riemann-surface structure for which the piece maps are local charts.

Solution. Use the images of the original domains as charts. On overlaps their transition maps are exactly the prescribed biholomorphisms, so the atlas is compatible and defines the complex structure. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 25.17 (Sphere gluing proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Gluing two copies of $\mathbb{C}$ by $w = 1/z$ on $\mathbb{C}^*$ produces a compact Riemann surface biholomorphic to $\widehat{\mathbb{C}}$.

Solution. Map one chart to finite complex numbers and the origin of the second chart to infinity. The transition relation is precisely the standard coordinate change at infinity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 25.18 (Cocycle consistency proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For triple overlaps, transition maps must satisfy $g_{ij} \circ g_{jk} = g_{ik}$.

Solution. A point represented through three charts must have the same coordinate identification independent of the path through the overlap graph; the cocycle condition is exactly that independence. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 25.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Gluing constructs Riemann surfaces by identifying open pieces through holomorphic transition maps. This is the global counterpart of defining a manifold from compatible charts. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

25.13 Exercises with complete solutions

Exercise 25.20. State and apply the central principle behind Gluing data. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Start with complex domains and specify identifications between open subsets. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 25.21. State and apply the central principle behind Transition maps. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Identifications must be biholomorphic and satisfy inverse and cocycle compatibility. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 25.22. State and apply the central principle behind Quotient space. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The disjoint union modulo the identifications becomes the candidate surface. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 25.23. State and apply the central principle behind Hausdorff condition. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Gluing must be controlled so the quotient remains Hausdorff. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 25.24. State and apply the central principle behind Charts from pieces. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Each original domain descends locally to a coordinate chart on the quotient. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 25.25. State and apply the central principle behind Sphere from two charts. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Two copies of the plane glued by $w = 1/z$ on punctured regions produce the Riemann sphere. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.26. State and apply the central principle behind Algebraic curves. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Local branches of an algebraic relation can be glued into a smooth Riemann surface. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 25.27. State and apply the central principle behind Cut-and-paste constructions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Copies of slit planes or polygons can be glued to produce root surfaces and tori. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 26

Compactification and Genus

Compactification adds missing points in a way compatible with the complex structure, while genus records the basic topological type of a compact connected Riemann surface.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

26.1 Adding infinity

The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$.

26.2 Puncture filling

A puncture can sometimes be filled when the local complex structure extends across it.

26.3 Compact algebraic curves

Affine algebraic curves often acquire points at infinity in their projective compactification.

26.4 Genus

For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$.

26.5 Sphere and torus

The sphere has genus zero; complex tori have genus one.

26.6 Meromorphic functions

On compact Riemann surfaces, nonconstant holomorphic functions to $\mathbb{C}$ cannot exist, but meromorphic functions correspond to holomorphic maps to the sphere.

26.7 Branched-cover viewpoint

Maps to the sphere reveal genus through their degree and ramification.

26.8 Riemann–Hurwitz formula

For finite branched covers, Euler characteristics differ by a total ramification correction.

26.9 Compactness consequences

Holomorphic functions, divisors, and differential forms become globally constrained on compact surfaces.

26.10 Core structural results

Theorem 26.1 (Holomorphic functions on compact surfaces)**.** *Every holomorphic function from a compact connected Riemann surface to $\mathbb{C}$ is constant.*

Proof. The absolute value attains a maximum by compactness. In a local coordinate the maximum-modulus principle forces local constancy, and connectedness propagates it globally. □

Theorem 26.2 (Riemann–Hurwitz formula)**.** *For a degree-d holomorphic map $f : X \to Y$ of compact Riemann surfaces, $2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X}(e_p - 1)$.*

Proof. Triangulate the target with branch values among vertices and lift the triangulation. Faces and edges lift d-fold, while vertices lose exactly the total ramification multiplicity. Comparing Euler characteristics yields the formula. □

Theorem 26.3 (Genus of a double cover)**.** *A degree-two cover of the sphere branched simply at r points has genus $g = (r - 2)/2$.*

Proof. Apply Riemann–Hurwitz with $d = 2$, target genus zero, and total ramification r: $2g - 2 = -4 + r$. □

26.11 Worked examples

Example 26.4 (Adding infinity diagnostic)**.** Give a rigorous worked analysis of Adding infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 26.5 (Puncture filling diagnostic)**.** Give a rigorous worked analysis of Puncture filling. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A puncture can sometimes be filled when the local complex structure extends across it. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 26.6 (Compact algebraic curves diagnostic)**.** Give a rigorous worked analysis of Compact algebraic curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Affine algebraic curves often acquire points at infinity in their projective compactification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 26.7 (Genus diagnostic)**.** Give a rigorous worked analysis of Genus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

26.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 26.8 (Adding infinity diagnostic)**.** Give a rigorous worked analysis of Adding infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.9 (Puncture filling diagnostic)**.** Give a rigorous worked analysis of Puncture filling. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A puncture can sometimes be filled when the local complex structure extends across it. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.10 (Compact algebraic curves diagnostic)**.** Give a rigorous worked analysis of Compact algebraic curves. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Affine algebraic curves often acquire points at infinity in their projective compactification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.11 (Genus diagnostic)**.** Give a rigorous worked analysis of Genus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.12 (Sphere and torus diagnostic)**.** Give a rigorous worked analysis of Sphere and torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The sphere has genus zero; complex tori have genus one. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.13 (Meromorphic functions diagnostic)**.** Give a rigorous worked analysis of Meromorphic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. On compact Riemann surfaces, nonconstant holomorphic functions to $\mathbb{C}$ cannot exist, but meromorphic functions correspond to holomorphic maps to the sphere. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.14 (Branched-cover viewpoint diagnostic)**.** Give a rigorous worked analysis of Branched-cover viewpoint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Maps to the sphere reveal genus through their degree and ramification. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.15 (Riemann–Hurwitz formula diagnostic)**.** Give a rigorous worked analysis of Riemann–Hurwitz formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. For finite branched covers, Euler characteristics differ by a total ramification correction. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 26.16 (Holomorphic functions on compact surfaces proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Every holomorphic function from a compact connected Riemann surface to $\mathbb{C}$ is constant.

Solution. The absolute value attains a maximum by compactness. In a local coordinate the maximum-modulus principle forces local constancy, and connectedness propagates it globally. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 26.17 (Riemann–Hurwitz formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a degree-d holomorphic map $f : X \to Y$ of compact Riemann surfaces, $2g_X - 2 = d(2g_Y - 2) + \sum_{p\in X}(e_p - 1)$.

Solution. Triangulate the target with branch values among vertices and lift the triangulation. Faces and edges lift d-fold, while vertices lose exactly the total ramification multiplicity. Comparing Euler characteristics yields the formula. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 26.18 (Genus of a double cover proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A degree-two cover of the sphere branched simply at r points has genus $g = (r - 2)/2$.

Solution. Apply Riemann–Hurwitz with $d = 2$, target genus zero, and total ramification r: $2g - 2 = -4 + r$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 26.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Compactification adds missing points in a way compatible with the complex structure, while genus records the basic topological type of a compact connected Riemann surface. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

26.13 Exercises with complete solutions

Exercise 26.20. State and apply the central principle behind Adding infinity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The complex plane compactifies to the Riemann sphere by adding one point with coordinate $w = 1/z$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.21. State and apply the central principle behind Puncture filling. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A puncture can sometimes be filled when the local complex structure extends across it. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.22. State and apply the central principle behind Compact algebraic curves. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Affine algebraic curves often acquire points at infinity in their projective compactification. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.23. State and apply the central principle behind Genus. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For a compact orientable surface, genus counts handles and determines Euler characteristic $\chi = 2 - 2g$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.24. State and apply the central principle behind Sphere and torus. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The sphere has genus zero; complex tori have genus one. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.25. State and apply the central principle behind Meromorphic functions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. On compact Riemann surfaces, nonconstant holomorphic functions to $\mathbb{C}$ cannot exist, but meromorphic functions correspond to holomorphic maps to the sphere. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.26. State and apply the central principle behind Branched-cover viewpoint. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Maps to the sphere reveal genus through their degree and ramification. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 26.27. State and apply the central principle behind Riemann–Hurwitz formula. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. For finite branched covers, Euler characteristics differ by a total ramification correction. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Part VI

Elliptic Functions

Chapter 27

Lattices and Complex Tori

A lattice in the complex plane is a discrete rank-two subgroup, and quotienting by it produces a compact complex torus. This is the analytic model of every genus-one Riemann surface after choosing a base point.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

27.1 Lattices

A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$.

27.2 Fundamental parallelogram

Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications.

27.3 Complex torus

The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations.

27.4 Quotient projection

The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map.

27.5 Change of lattice basis

Replacing a lattice basis by an $SL_2(\mathbb{Z})$ basis does not change the lattice.

27.6 Normalized modulus

After scaling, a lattice can be represented by $\mathbb{Z} + \mathbb{Z}\tau$ with $\Im\tau > 0$.

27.7 Modular equivalence

Different τ values related by fractional linear $SL_2(\mathbb{Z})$ transformations describe isomorphic normalized tori.

27.8 Holomorphic differentials

The form dz descends to a nowhere-vanishing holomorphic differential on the torus.

27.9 Genus one

The torus has Euler characteristic zero and genus one.

27.10 Core structural results

Theorem 27.1 (Complex structure on the quotient)**.** *For a lattice Λ, the quotient $\mathbb{C}/\Lambda$ has a unique Riemann-surface structure making the projection locally biholomorphic.*

Proof. Choose disks small enough that distinct lattice translates are disjoint. Projection restricts to a homeomorphism on each such disk, and transition maps between lifted charts are translations, hence holomorphic. □

Theorem 27.2 (Compactness of the torus)**.** $\mathbb{C}/\Lambda$ *is compact.*

Proof. The closure of a fundamental parallelogram is compact and its projection surjects onto the quotient. A continuous image of a compact set is compact. □

Theorem 27.3 (Basis-change invariance)**.** *If $(\omega_1', \omega_2') = (a\omega_1 + b\omega_2, c\omega_1 + d\omega_2)$ with an integral matrix of determinant ± 1, then the generated lattice is unchanged.*

Proof. The new generators are integral combinations of the old, and invertibility over $\mathbb{Z}$ expresses the old generators as integral combinations of the new. □

27.11 Worked examples

Example 27.4 (Lattices diagnostic)**.** Give a rigorous worked analysis of Lattices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 27.5 (Fundamental parallelogram diagnostic)**.** Give a rigorous worked analysis of Fundamental parallelogram. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 27.6 (Complex torus diagnostic)**.** Give a rigorous worked analysis of Complex torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 27.7 (Quotient projection diagnostic)**.** Give a rigorous worked analysis of Quotient projection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

27.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 27.8 (Lattices diagnostic)**.** Give a rigorous worked analysis of Lattices. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.9 (Fundamental parallelogram diagnostic)**.** Give a rigorous worked analysis of Fundamental parallelogram. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.10 (Complex torus diagnostic)**.** Give a rigorous worked analysis of Complex torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.11 (Quotient projection diagnostic)**.** Give a rigorous worked analysis of Quotient projection. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.12 (Change of lattice basis diagnostic)**.** Give a rigorous worked analysis of Change of lattice basis. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Replacing a lattice basis by an $SL_2(\mathbb{Z})$ basis does not change the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 27.13 (Normalized modulus diagnostic)**.** Give a rigorous worked analysis of Normalized modulus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. After scaling, a lattice can be represented by $\mathbb{Z} + \mathbb{Z}\tau$ with $\Im\tau > 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 27.14 (Modular equivalence diagnostic)**.** Give a rigorous worked analysis of Modular equivalence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Different τ values related by fractional linear $SL_2(\mathbb{Z})$ transformations describe isomorphic normalized tori. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 27.15 (Holomorphic differentials diagnostic)**.** Give a rigorous worked analysis of Holomorphic differentials. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The form dz descends to a nowhere-vanishing holomorphic differential on the torus. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 27.16 (Complex structure on the quotient proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For a lattice Λ, the quotient $\mathbb{C}/\Lambda$ has a unique Riemann-surface structure making the projection locally biholomorphic.

Solution. Choose disks small enough that distinct lattice translates are disjoint. Projection restricts to a homeomorphism on each such disk, and transition maps between lifted charts are translations, hence holomorphic. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 27.17 (Compactness of the torus proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: $\mathbb{C}/\Lambda$ is compact.

Solution. The closure of a fundamental parallelogram is compact and its projection surjects onto the quotient. A continuous image of a compact set is compact. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 27.18 (Basis-change invariance proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $(\omega_1', \omega_2') = (a\omega_1 + b\omega_2, c\omega_1 + d\omega_2)$ with an integral matrix of determinant ± 1, then the generated lattice is unchanged.

Solution. The new generators are integral combinations of the old, and invertibility over $\mathbb{Z}$ expresses the old generators as integral combinations of the new. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 27.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. A lattice in the complex plane is a discrete rank-two subgroup, and quotienting by it produces a compact complex torus. This is the analytic model of every genus-one Riemann surface after choosing a base point. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

27.13 Exercises with complete solutions

Exercise 27.20. State and apply the central principle behind Lattices. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A lattice has form $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ with $\omega_1/\omega_2 \notin \mathbb{R}$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.21. State and apply the central principle behind Fundamental parallelogram. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Every complex number is congruent modulo the lattice to a point in a chosen fundamental parallelogram, with boundary identifications. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.22. State and apply the central principle behind Complex torus. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The quotient $\mathbb{C}/\Lambda$ inherits a compact Riemann-surface structure from translations. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.23. State and apply the central principle behind Quotient projection. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The map $\pi : \mathbb{C} \to \mathbb{C}/\Lambda$ is a holomorphic covering map. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.24. State and apply the central principle behind Change of lattice basis. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Replacing a lattice basis by an $SL_2(\mathbb{Z})$ basis does not change the lattice. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.25. State and apply the central principle behind Normalized modulus. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. After scaling, a lattice can be represented by $\mathbb{Z}+\mathbb{Z}\tau$ with $\Im\tau > 0$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.26. State and apply the central principle behind Modular equivalence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Different τ values related by fractional linear $SL_2(\mathbb{Z})$ transformations describe isomorphic normalized tori. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 27.27. State and apply the central principle behind Holomorphic differentials. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The form dz descends to a nowhere-vanishing holomorphic differential on the torus. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 28

Elliptic Functions

Elliptic functions are meromorphic functions periodic with respect to a rank-two lattice. Compactness of the associated torus forces strong balance laws for zeros, poles, and residues.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

28.1 Double periodicity

An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$.

28.2 Descent to the torus

Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane.

28.3 Poles in a period cell

A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant.

28.4 Residue balance

The sum of residues in a fundamental parallelogram is zero.

28.5 Zero-pole balance

The number of zeros equals the number of poles in a period parallelogram, counted with multiplicity.

28.6 Abel-type position constraint

The sum of zeros minus the sum of poles is a lattice element, counted with multiplicity.

28.7 Constructing elliptic functions

Periodic principal parts must satisfy residue compatibility before a global elliptic function can exist.

28.8 Even and odd examples

The Weierstrass $\wp$-function is even; its derivative is odd.

28.9 Function-field preview

For a fixed lattice, elliptic functions are generated algebraically by $\wp$ and $\wp'$.

28.10 Core structural results

Theorem 28.1 (Residue sum theorem)**.** *The sum of residues of an elliptic function in a fundamental parallelogram containing no poles on its boundary is zero.*

Proof. Integrate around the boundary. Opposite edges cancel because the function is periodic and the orientations are opposite. The residue theorem then forces the interior residue sum to vanish. □

Theorem 28.2 (Equal zero and pole count)**.** *A nonzero elliptic function has equally many zeros and poles in a fundamental parallelogram, counted with multiplicity.*

Proof. Apply the argument principle to f'/f. This quotient is elliptic, so opposite-edge contour integrals cancel, leaving zero; hence zeros minus poles equals zero. □

Theorem 28.3 (No single simple pole)**.** *There is no elliptic function with exactly one simple pole modulo the lattice.*

Proof. Such a function would have one nonzero residue in a fundamental parallelogram, contradicting the residue-sum theorem. □

28.11 Worked examples

Example 28.4 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 28.5 (Descent to the torus diagnostic)**.** Give a rigorous worked analysis of Descent to the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 28.6 (Poles in a period cell diagnostic)**.** Give a rigorous worked analysis of Poles in a period cell. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 28.7 (Residue balance diagnostic)**.** Give a rigorous worked analysis of Residue balance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The sum of residues in a fundamental parallelogram is zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

28.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 28.8 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. An elliptic function satisfies $f(z+\omega) = f(z)$ for every lattice period $\omega \in \Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.9 (Descent to the torus diagnostic)**.** Give a rigorous worked analysis of Descent to the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.10 (Poles in a period cell diagnostic)**.** Give a rigorous worked analysis of Poles in a period cell. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.11 (Residue balance diagnostic)**.** Give a rigorous worked analysis of Residue balance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The sum of residues in a fundamental parallelogram is zero. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.12 (Zero-pole balance diagnostic)**.** Give a rigorous worked analysis of Zero-pole balance. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The number of zeros equals the number of poles in a period parallelogram, counted with multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.13 (Abel-type position constraint diagnostic)**.** Give a rigorous worked analysis of Abel-type position constraint. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The sum of zeros minus the sum of poles is a lattice element, counted with multiplicity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.14 (Constructing elliptic functions diagnostic)**.** Give a rigorous worked analysis of Constructing elliptic functions. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Periodic principal parts must satisfy residue compatibility before a global elliptic function can exist. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.15 (Even and odd examples diagnostic)**.** Give a rigorous worked analysis of Even and odd examples. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The Weierstrass $\wp$-function is even; its derivative is odd. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 28.16 (Residue sum theorem proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The sum of residues of an elliptic function in a fundamental parallelogram containing no poles on its boundary is zero.

Solution. Integrate around the boundary. Opposite edges cancel because the function is periodic and the orientations are opposite. The residue theorem then forces the interior residue sum to vanish. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 28.17 (Equal zero and pole count proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: A nonzero elliptic function has equally many zeros and poles in a fundamental parallelogram, counted with multiplicity.

Solution. Apply the argument principle to f'/f. This quotient is elliptic, so opposite-edge contour integrals cancel, leaving zero; hence zeros minus poles equals zero. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 28.18 (No single simple pole proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: There is no elliptic function with exactly one simple pole modulo the lattice.

Solution. Such a function would have one nonzero residue in a fundamental parallelogram, contradicting the residue-sum theorem. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 28.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Elliptic functions are meromorphic functions periodic with respect to a rank-two lattice. Compactness of the associated torus forces strong balance laws for zeros, poles, and residues. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. □

28.13 Exercises with complete solutions

Exercise 28.20. State and apply the central principle behind Double periodicity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. An elliptic function satisfies $f(z + \omega) = f(z)$ for every lattice period $\omega \in \Lambda$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.21. State and apply the central principle behind Descent to the torus. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Elliptic functions are exactly meromorphic functions on $\mathbb{C}/\Lambda$ pulled back to the plane. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.22. State and apply the central principle behind Poles in a period cell. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. A nonconstant elliptic function must have poles because a holomorphic function on the compact torus would be constant. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.23. State and apply the central principle behind Residue balance. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The sum of residues in a fundamental parallelogram is zero. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.24. State and apply the central principle behind Zero-pole balance. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The number of zeros equals the number of poles in a period parallelogram, counted with multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.25. State and apply the central principle behind Abel-type position constraint. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The sum of zeros minus the sum of poles is a lattice element, counted with multiplicity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.26. State and apply the central principle behind Constructing elliptic functions. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Periodic principal parts must satisfy residue compatibility before a global elliptic function can exist. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 28.27. State and apply the central principle behind Even and odd examples. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The Weierstrass $\wp$-function is even; its derivative is odd. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 29

The Weierstrass $\wp$-Function

The Weierstrass $\wp$-function is the canonical elliptic function attached to a lattice. Its carefully regularized lattice sum gives a doubly periodic even function with one double pole per period cell and an algebraic differential equation.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$

29.1 Regularized lattice sum

Define $\wp(z) = z^{-2} + \sum_{\omega \in \Lambda \setminus \{0\}} [(z-\omega)^{-2} - \omega^{-2}]$.

29.2 Normal convergence

The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice.

29.3 Evenness

Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$.

29.4 Double periodicity

The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic.

29.5 Pole structure

Modulo the lattice, $\wp$ has one double pole at the origin with zero residue.

29.6 Derivative

The derivative $\wp'$ is odd and has a triple pole at the lattice points.

29.7 Half-period values

At nonzero half-periods, $\wp'$ vanishes; their $\wp$ values are the roots e_1, e_2, e_3 of the cubic relation.

29.8 Differential equation

There are lattice invariants g_2, g_3 with $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

29.9 Second derivative relation

Differentiating the cubic gives $\wp'' = 6\wp^2 - g_2/2$.

29.10 Uniformization preview

The map $z \mapsto (\wp(z), \wp'(z))$ parametrizes a nonsingular cubic curve when the discriminant is nonzero.

29.11 Core structural results

Theorem 29.1 (Normal convergence of the Weierstrass series)**.** *The defining series for $\wp$ converges normally on compact subsets of $\mathbb{C} \setminus \Lambda$.*

Proof. For bounded z and large $|\omega|$, expand $(z-\omega)^{-2} - \omega^{-2} = O(|z|/|\omega|^3)$. The lattice sum $\sum_{\omega \neq 0} |\omega|^{-3}$ converges, so the Weierstrass M-test applies away from lattice points. □

Theorem 29.2 (Elliptic differential equation)**.** *There exist constants $g_2 = 60\sum_{\omega\neq 0}\omega^{-4}$ and $g_3 = 140\sum_{\omega\neq 0}\omega^{-6}$ such that $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.*

Proof. Compare Laurent expansions at zero. The difference $(\wp')^2 - 4\wp^3 + g_2\wp + g_3$ is elliptic and has its principal part cancelled, hence is entire on the torus and therefore constant. The chosen coefficients make the constant zero. □

Theorem 29.3 (Half-period critical points)**.** *If $2a \in \Lambda$ but $a \notin \Lambda$, then $\wp'(a) = 0$.*

Proof. Because $\wp'$ is odd and periodic, $\wp'(a) = \wp'(a - 2a) = \wp'(-a) = -\wp'(a)$. □

Theorem 29.4 (Degree-two map to the sphere)**.** *The function $\wp$ induces a degree-two meromorphic map $\mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$, branched at the three nonzero half-periods and the origin.*

Proof. Generically $\wp(z) = \wp(-z)$ gives two points in a fiber. Fixed points of the involution $z \mapsto -z$ modulo the lattice are exactly the four half-period classes, where the local multiplicity rises to two. □

29.12 Worked examples

Example 29.5 (Regularized lattice sum diagnostic)**.** Give a rigorous worked analysis of Regularized lattice sum. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Define $\wp(z) = z^{-2} + \sum_{\omega\in\Lambda\setminus\{0\}}[(z-\omega)^{-2} - \omega^{-2}]$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 29.6 (Normal convergence diagnostic)**.** Give a rigorous worked analysis of Normal convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 29.7 (Evenness diagnostic)**.** Give a rigorous worked analysis of Evenness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 29.8 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

29.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 29.9 (Regularized lattice sum diagnostic)**.** Give a rigorous worked analysis of Regularized lattice sum. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Define $\wp(z) = z^{-2} + \sum_{\omega\in\Lambda\setminus\{0\}}[(z-\omega)^{-2} - \omega^{-2}]$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.10 (Normal convergence diagnostic)**.** Give a rigorous worked analysis of Normal convergence. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.11 (Evenness diagnostic)**.** Give a rigorous worked analysis of Evenness. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.12 (Double periodicity diagnostic)**.** Give a rigorous worked analysis of Double periodicity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.13 (Pole structure diagnostic)**.** Give a rigorous worked analysis of Pole structure. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Modulo the lattice, $\wp$ has one double pole at the origin with zero residue. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.14 (Derivative diagnostic)**.** Give a rigorous worked analysis of Derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The derivative $\wp'$ is odd and has a triple pole at the lattice points. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.15 (Half-period values diagnostic)**.** Give a rigorous worked analysis of Half-period values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. At nonzero half-periods, $\wp'$ vanishes; their $\wp$ values are the roots e_1, e_2, e_3 of the cubic relation. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 29.16 (Normal convergence of the Weierstrass series proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The defining series for $\wp$ converges normally on compact subsets of $\mathbb{C} \setminus \Lambda$.

Solution. For bounded z and large $|\omega|$, expand $(z-\omega)^{-2} - \omega^{-2} = O(|z|/|\omega|^3)$. The lattice sum $\sum_{\omega\neq 0} |\omega|^{-3}$ converges, so the Weierstrass M-test applies away from lattice points. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.17 (Elliptic differential equation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: There exist constants $g_2 = 60\sum_{\omega\neq 0} \omega^{-4}$ and $g_3 = 140\sum_{\omega\neq 0} \omega^{-6}$ such that $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

Solution. Compare Laurent expansions at zero. The difference $(\wp')^2 - 4\wp^3 + g_2\wp + g_3$ is elliptic and has its principal part cancelled, hence is entire on the torus and therefore constant. The chosen coefficients make the constant zero. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.18 (Half-period critical points proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If $2a \in \Lambda$ but $a \notin \Lambda$, then $\wp'(a) = 0$.

Solution. Because $\wp'$ is odd and periodic, $\wp'(a) = \wp'(a-2a) = \wp'(-a) = -\wp'(a)$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 29.19 (Degree-two map to the sphere proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The function $\wp$ induces a degree-two meromorphic map $\mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$, branched at the three nonzero half-periods and the origin.

Solution. Generically $\wp(z) = \wp(-z)$ gives two points in a fiber. Fixed points of the involution $z \mapsto -z$ modulo the lattice are exactly the four half-period classes, where the local multiplicity rises to two. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 29.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Weierstrass $\wp$-function is the canonical elliptic function attached to a lattice. Its carefully regularized lattice sum gives a doubly periodic even function with one double pole per period cell and an algebraic differential equation. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

29.14 Exercises with complete solutions

Exercise 29.21. State and apply the central principle behind Regularized lattice sum. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Define $\wp(z) = z^{-2} + \sum_{\omega\in\Lambda\setminus\{0\}}[(z-\omega)^{-2} - \omega^{-2}]$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.22. State and apply the central principle behind Normal convergence. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The subtraction of ω^{-2} improves convergence so the series converges normally on compact sets avoiding the lattice. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.23. State and apply the central principle behind Evenness. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Pairing lattice points ω and $-\omega$ shows $\wp(-z) = \wp(z)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.24. State and apply the central principle behind Double periodicity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The derivative $\wp'$ is manifestly lattice periodic, and comparison constants show $\wp$ itself is elliptic. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 29.25. State and apply the central principle behind Pole structure. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Modulo the lattice, $\wp$ has one double pole at the origin with zero residue. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 29.26. State and apply the central principle behind Derivative. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The derivative $\wp'$ is odd and has a triple pole at the lattice points. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 29.27. State and apply the central principle behind Half-period values. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. At nonzero half-periods, $\wp'$ vanishes; their $\wp$ values are the roots e_1, e_2, e_3 of the cubic relation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 29.28. State and apply the central principle behind Differential equation. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. There are lattice invariants g_2, g_3 with $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 30

Addition Formulas

Addition formulas express the group structure of the complex torus directly in the values of $\wp$ and $\wp'$. They are the analytic form of the chord-and-tangent law on the associated cubic curve.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

30.1 Translation on the torus

Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$.

30.2 Need for an addition law

Since $\wp$ is even, $\wp(u + v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data.

30.3 Weierstrass addition formula

A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u + v)$.

30.4 Duplication formula

Taking $v \to u$ yields the tangent or duplication formula.

30.5 Subtraction formula

Replacing v by $-v$ gives a companion formula for $\wp(u - v)$.

30.6 Pole comparison proof

Differences of candidate expressions can be shown constant by comparing elliptic principal parts.

30.7 Cubic geometry

The slope term in the addition formula is the slope of the chord joining points on the Weierstrass cubic.

30.8 Special values

Half-periods and torsion points simplify the formulas because $\wp'$ may vanish.

30.9 Division phenomena

Iterating addition generates rational functions describing multiplication-by-n on the elliptic curve.

30.10 Core structural results

Theorem 30.1 (Weierstrass addition formula)**.** *For $u, v, u \pm v \notin \Lambda$ and $\wp(u) \neq \wp(v)$, $\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2$.*

Proof. As a function of one variable, compare both sides after shifting by the other argument. Their poles and principal parts agree; the difference is an elliptic function without poles, hence constant. Evaluate at a convenient limiting point to determine the constant. □

Theorem 30.2 (Duplication formula)**.** *The limit $v \to u$ in the addition formula gives $\wp(2u) = -2\wp(u) + \frac{1}{4}(\wp''(u)/\wp'(u))^2$ whenever the displayed quotient is defined.*

Proof. Apply l'Hôpital's rule to the chord-slope quotient as $v \to u$. □

Theorem 30.3 (Chord interpretation)**.** *Under $z \mapsto (\wp(z), \wp'(z))$, the analytic addition formula agrees with the chord-and-tangent law on $y^2 = 4x^3 - g_2x - g_3$.*

Proof. The squared chord slope determines the third intersection's x-coordinate by Vieta's formula; reflection in the x-axis changes the sign of y, matching the torus inverse $z \mapsto -z$. □

30.11 Worked examples

Example 30.4 (Translation on the torus diagnostic)**.** Give a rigorous worked analysis of Translation on the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 30.5 (Need for an addition law diagnostic)**.** Give a rigorous worked analysis of Need for an addition law. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 30.6 (Weierstrass addition formula diagnostic)**.** Give a rigorous worked analysis of Weierstrass addition formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 30.7 (Duplication formula diagnostic)**.** Give a rigorous worked analysis of Duplication formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. Taking $v \to u$ yields the tangent or duplication formula. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

30.12 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 30.8 (Translation on the torus diagnostic)**.** Give a rigorous worked analysis of Translation on the torus. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.9 (Need for an addition law diagnostic)**.** Give a rigorous worked analysis of Need for an addition law. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.10 (Weierstrass addition formula diagnostic)**.** Give a rigorous worked analysis of Weierstrass addition formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.11 (Duplication formula diagnostic)**.** Give a rigorous worked analysis of Duplication formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Taking $v \to u$ yields the tangent or duplication formula. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.12 (Subtraction formula diagnostic)**.** Give a rigorous worked analysis of Subtraction formula. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Replacing v by $-v$ gives a companion formula for $\wp(u-v)$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.13 (Pole comparison proof diagnostic)**.** Give a rigorous worked analysis of Pole comparison proof. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Differences of candidate expressions can be shown constant by comparing elliptic principal parts. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.14 (Cubic geometry diagnostic)**.** Give a rigorous worked analysis of Cubic geometry. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The slope term in the addition formula is the slope of the chord joining points on the Weierstrass cubic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.15 (Special values diagnostic)**.** Give a rigorous worked analysis of Special values. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Half-periods and torsion points simplify the formulas because $\wp'$ may vanish. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. ☐

Problem 30.16 (Weierstrass addition formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For $u, v, u \pm v \notin \Lambda$ and $\wp(u) \neq \wp(v)$, $\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u)-\wp'(v)}{\wp(u)-\wp(v)}\right)^2$.

Solution. As a function of one variable, compare both sides after shifting by the other argument. Their poles and principal parts agree; the difference is an elliptic function without poles, hence constant. Evaluate at a convenient limiting point to determine the constant. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 30.17 (Duplication formula proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The limit $v \to u$ in the addition formula gives $\wp(2u) = -2\wp(u) + \frac{1}{4}(\wp''(u)/\wp'(u))^2$ whenever the displayed quotient is defined.

Solution. Apply l'Hôpital's rule to the chord-slope quotient as $v \to u$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 30.18 (Chord interpretation proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Under $z \mapsto (\wp(z), \wp'(z))$, the analytic addition formula agrees with the chord-and-tangent law on $y^2 = 4x^3 - g_2x - g_3$.

Solution. The squared chord slope determines the third intersection's x-coordinate by Vieta's formula; reflection in the x-axis changes the sign of y, matching the torus inverse $z \mapsto -z$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 30.19 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. Addition formulas express the group structure of the complex torus directly in the values of $\wp$ and $\wp'$. They are the analytic form of the chord-and-tangent law on the associated cubic curve. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

30.13 Exercises with complete solutions

Exercise 30.20. State and apply the central principle behind Translation on the torus. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Addition of complex arguments modulo the lattice is the group law on $\mathbb{C}/\Lambda$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.21. State and apply the central principle behind Need for an addition law. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Since $\wp$ is even, $\wp(u+v)$ cannot be expressed from $\wp(u), \wp(v)$ alone without derivative data. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.22. State and apply the central principle behind Weierstrass addition formula. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. A rational expression in $\wp(u), \wp(v), \wp'(u), \wp'(v)$ gives $\wp(u+v)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.23. State and apply the central principle behind Duplication formula. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Taking $v \to u$ yields the tangent or duplication formula. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.24. State and apply the central principle behind Subtraction formula. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. Replacing v by $-v$ gives a companion formula for $\wp(u-v)$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 30.25. State and apply the central principle behind Pole comparison proof. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Differences of candidate expressions can be shown constant by comparing elliptic principal parts. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 30.26. State and apply the central principle behind Cubic geometry. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The slope term in the addition formula is the slope of the chord joining points on the Weierstrass cubic. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 30.27. State and apply the central principle behind Special values. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Half-periods and torsion points simplify the formulas because $\wp'$ may vanish. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.

Chapter 31

Elliptic Curves as Riemann Surfaces

The Weierstrass parametrization identifies a complex torus with a smooth projective cubic curve. Analytic addition on the torus becomes the algebraic group law of an elliptic curve.

Learning goals

The reader should be able to state the definitions precisely, prove the core results, choose valid contours or local expansions, and diagnose which domain and regularity hypotheses are essential.

Conceptual roadmap

$$\boxed{\text{local holomorphic structure} \longrightarrow \text{integral or series representation} \longrightarrow \text{global consequence.}}$$

31.1 Weierstrass cubic

The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$.

31.2 Parametrization

The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity.

31.3 Point at infinity

The unique point at infinity is the identity of the cubic group law.

31.4 Injectivity modulo sign and derivative

The pair $(\wp, \wp')$ separates torus points modulo the lattice.

31.5 Surjectivity

Every point of the smooth cubic arises from a torus point.

31.6 Biholomorphism

The extended Weierstrass map is a holomorphic bijection between compact Riemann surfaces and hence a biholomorphism.

31.7 Group law

Addition of arguments corresponds to the chord-and-tangent group law on the cubic.

31.8 Invariant differential

The form dx/y pulls back to dz up to the normalization determined by the chosen cubic equation.

31.9 Moduli viewpoint

Changing the lattice by scaling changes g_2, g_3 with weights and preserves the modular invariant j.

31.10 Genus-one synthesis

Complex tori, smooth cubic curves, elliptic functions, and genus-one Riemann surfaces with a base point describe the same analytic geometry.

31.11 Core structural results

Theorem 31.1 (Weierstrass map lies on the cubic)**.** *For every $z \notin \Lambda$, the pair $(\wp(z), \wp'(z))$ satisfies $y^2 = 4x^3 - g_2x - g_3$.*

Proof. This is exactly the differential equation for $\wp$. □

Theorem 31.2 (Extension at the origin)**.** *The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends holomorphically at the lattice class of zero to the projective point at infinity.*

Proof. Using $\wp(z) = z^{-2} + O(z^2)$ and $\wp'(z) = -2z^{-3} + O(z)$, rescale homogeneous coordinates by a suitable power of z to obtain a finite nonzero projective limit equal to the unique point at infinity. □

Theorem 31.3 (Torus-cubic biholomorphism)**.** *If the discriminant is nonzero, the extended Weierstrass map $\mathbb{C}/\Lambda \to E$ is a biholomorphism onto the smooth cubic E.*

Proof. The map is nonconstant and holomorphic between compact Riemann surfaces. The pair $\wp, \wp'$ separates points modulo the lattice, giving injectivity; compactness makes the image closed, while nonconstant holomorphic maps are open, so the image is all of the connected cubic. A holomorphic bijection between compact Riemann surfaces has holomorphic inverse. □

Theorem 31.4 (Compatibility of group laws)**.** *Under the Weierstrass biholomorphism, addition on $\mathbb{C}/\Lambda$ corresponds to the cubic chord-and-tangent group law.*

Proof. The addition formula computes the coordinates of the image of $u + v$ and matches the algebraic chord construction; the identity is the lattice origin mapped to infinity and inversion is $z \mapsto -z$, which changes the sign of $\wp'$. □

31.12 Worked examples

Example 31.5 (Weierstrass cubic diagnostic)**.** Give a rigorous worked analysis of Weierstrass cubic. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 31.6 (Parametrization diagnostic)**.** Give a rigorous worked analysis of Parametrization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 31.7 (Point at infinity diagnostic)**.** Give a rigorous worked analysis of Point at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The unique point at infinity is the identity of the cubic group law. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

Example 31.8 (Injectivity modulo sign and derivative diagnostic)**.** Give a rigorous worked analysis of Injectivity modulo sign and derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion. The pair $(\wp, \wp')$ separates torus points modulo the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone.

31.13 Solved dossiers

The dossiers are canonical solved problems. The provenance ledger separately records which retained corpus rules guided each topic and which dossiers were newly designed.

Problem 31.9 (Weierstrass cubic diagnostic)**.** Give a rigorous worked analysis of Weierstrass cubic. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.10 (Parametrization diagnostic)**.** Give a rigorous worked analysis of Parametrization. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.11 (Point at infinity diagnostic)**.** Give a rigorous worked analysis of Point at infinity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The unique point at infinity is the identity of the cubic group law. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.12 (Injectivity modulo sign and derivative diagnostic)**.** Give a rigorous worked analysis of Injectivity modulo sign and derivative. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The pair $(\wp, \wp')$ separates torus points modulo the lattice. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.13 (Surjectivity diagnostic)**.** Give a rigorous worked analysis of Surjectivity. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Every point of the smooth cubic arises from a torus point. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.14 (Biholomorphism diagnostic)**.** Give a rigorous worked analysis of Biholomorphism. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. The extended Weierstrass map is a holomorphic bijection between compact Riemann surfaces and hence a biholomorphism. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.15 (Group law diagnostic)**.** Give a rigorous worked analysis of Group law. State the relevant holomorphic, contour, series, or singularity hypothesis and derive the central conclusion.

Solution. Addition of arguments corresponds to the chord-and-tangent group law on the cubic. The decisive step is to use the complex-analytic structure globally rather than reason from real-variable intuition alone. □

Problem 31.16 (Weierstrass map lies on the cubic proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: For every $z \notin \Lambda$, the pair $(\wp(z), \wp'(z))$ satisfies $y^2 = 4x^3 - g_2x - g_3$.

Solution. This is exactly the differential equation for $\wp$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 31.17 (Extension at the origin proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends holomorphically at the lattice class of zero to the projective point at infinity.

Solution. Using $\wp(z) = z^{-2} + O(z^2)$ and $\wp'(z) = -2z^{-3} + O(z)$, rescale homogeneous coordinates by a suitable power of z to obtain a finite nonzero projective limit equal to the unique point at infinity. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. □

Problem 31.18 (Torus-cubic biholomorphism proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: If the discriminant is nonzero, the extended Weierstrass map $\mathbb{C}/\Lambda \to E$ is a biholomorphism onto the smooth cubic E.

Solution. The map is nonconstant and holomorphic between compact Riemann surfaces. The pair $\wp, \wp'$ separates points modulo the lattice, giving injectivity; compactness makes the image closed, while nonconstant holomorphic maps are open, so the image is all of the connected cubic. A holomorphic bijection between compact Riemann surfaces has holomorphic inverse. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 31.19 (Compatibility of group laws proof dossier)**.** Prove the following theorem-level statement and identify the essential hypothesis: Under the Weierstrass biholomorphism, addition on $\mathbb{C}/\Lambda$ corresponds to the cubic chord-and-tangent group law.

Solution. The addition formula computes the coordinates of the image of $u+v$ and matches the algebraic chord construction; the identity is the lattice origin mapped to infinity and inversion is $z \mapsto -z$, which changes the sign of $\wp'$. This identifies the mechanism that makes the complex-analytic conclusion stronger than its real-variable analogue. ☐

Problem 31.20 (Chapter synthesis)**.** Connect the definitions and principal theorems of this chapter into one reusable complex-analysis strategy.

Solution. The Weierstrass parametrization identifies a complex torus with a smooth projective cubic curve. Analytic addition on the torus becomes the algebraic group law of an elliptic curve. The reusable strategy is to identify the analytic domain, choose the correct local expansion or contour representation, apply the structural theorem, and then audit singularities and topology. ☐

31.14 Exercises with complete solutions

Exercise 31.21. State and apply the central principle behind Weierstrass cubic. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The equation $y^2 = 4x^3 - g_2x - g_3$ defines a nonsingular cubic when $g_2^3 - 27g_3^2 \neq 0$. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 31.22. State and apply the central principle behind Parametrization. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The map $z \mapsto [\wp(z) : \wp'(z) : 1]$ extends across the lattice point to the point at infinity. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 31.23. State and apply the central principle behind Point at infinity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The unique point at infinity is the identity of the cubic group law. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. ☐

Exercise 31.24. State and apply the central principle behind Injectivity modulo sign and derivative. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. ☐

Solution. The pair $(\wp, \wp')$ separates torus points modulo the lattice. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.25. State and apply the central principle behind Surjectivity. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Every point of the smooth cubic arises from a torus point. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.26. State and apply the central principle behind Biholomorphism. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The extended Weierstrass map is a holomorphic bijection between compact Riemann surfaces and hence a biholomorphism. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.27. State and apply the central principle behind Group law. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. Addition of arguments corresponds to the chord-and-tangent group law on the cubic. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Exercise 31.28. State and apply the central principle behind Invariant differential. Give one concrete consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. □

Solution. The form dx/y pulls back to dz up to the normalization determined by the chosen cubic equation. The same principle can then be reused in later contour, residue, conformal, or Riemann-surface arguments. □

Chapter summary

The chapter now forms part of the canonical complex-analysis chain from holomorphicity through residues, global continuation, and Riemann surfaces.